

The Floor Beneath the Portfolio

Public Pension Design and the All-Equity Lifecycle Portfolio

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Abstract

A lifecycle investor holding only equities, split between domestic and international markets, is said to dominate the age-declining glide path embedded in target-date funds. We show that this prescription is a property of the public pension the investor retires onto, not of the return process. On a 16-country panel of real returns spanning 1890–2020, simulated with a calendar-joint block bootstrap, the all-equity portfolio leads the target-date fund by 5.73% in certainty-equivalent retirement consumption under the United States' earnings-related schedule. Replacing that schedule with Australia's — a means-tested Age Pension alongside a compulsory 12% Superannuation Guarantee — reverses the ordering to -6.42%, and the target-date fund wins.

The mechanism is not the means test's taper. This household holds 32.1 times average earnings against a cut-off of 6.8, and 98% of simulated households are past the test before it is applied. What the means test removes is an unconditional real annuity, and with it the floor under bad outcomes: mean retirement consumption *rises* +27% while the fifth percentile *falls* -27%. Paying the same flat pension unconditionally, means-testing nothing, leaves the ranking intact at +6.33%, which isolates the test rather than the pension's level.

A floor need not come from a pension. Under an amortisation withdrawal rule — which divides the balance by the years remaining and therefore cannot deplete it — ruin falls to zero and the ordering reverses again: scored at each system's own best assumed return (4% for the United States, 7% for Australia) the Australian household leads by +11.3%. The all-equity prescription is therefore conditional on two institutions at once, the pension and the drawdown rule, and is silent about a household that has neither a guaranteed income nor a rule that cannot run out.

Keywords: lifecycle asset allocation; public pension design; means testing; target-date funds; certainty-equivalent consumption; decumulation. **JEL:** G11, G51, H55, D14.

1. Introduction

The default investment vehicle of the modern retirement system is the target-date fund, and it embodies one proposition: that an investor should hold less equity as they age. Anarkulova, Cederburg and O'Doherty (2023) challenge it directly, showing that when returns are drawn in blocks from the international historical record rather than i.i.d. from post-war US data, a fixed all-equity portfolio beats the glide path on almost every metric an investor would care about.

That result is derived for an investor who receives the United States' Social Security: an earnings-related benefit, paid whatever else the retiree owns. This paper asks what the prescription becomes for an investor who does not.

1.1 What we find

The ordering reverses under an asset-tested pension. The all-equity lead of 5.73% becomes -6.42% when the American schedule is replaced by Australia's, and the de-risking glide path takes first place. Nothing about the return panel changes between those two runs; the objective function does.

The mechanism is the floor, not the taper. A means test is usually discussed as an implicit tax on wealth, and Australia's is steep enough to qualify — a dollar of assessable assets costs 7.8% of pension a year, more than domestic equity earns on average in this panel. But that reading does not survive checking where the household stands: it is past the cut-off before the test is applied, so the taper reaches it only in the left tail, after a portfolio has already fallen. What the means test removes is the unconditional annuity itself. A retiree standing on a guaranteed floor can carry the equity tail; one standing on their portfolio alone cannot, and a risk-averse objective prices that difference heavily.

A floor need not come from a pension. The reversal is conditional on a withdrawal rule that can run out. Under an amortisation rule the balance is divided by the years remaining, ruin falls to zero, and the household manufactures its own floor out of the larger portfolio compulsory saving bought it. The ordering then reverses a second time. Which means the all-equity prescription is conditional on two institutions rather than one.

1.2 Relation to the literature

The lifecycle portfolio-choice literature since Cocco, Gomes and Maenhout (2005) and Gomes and Michaelides (2005) treats public pension income as a bond-like endowment that crowds fixed income out of the financial portfolio. That is right for an earnings-related schedule. It is the wrong intuition for an asset-tested one, where the endowment is withdrawn precisely as the portfolio grows, and where — as Hubbard, Skinner and Zeldes (1995) show for saving — the binding feature is the floor the transfer provides rather than its expected level. Dahlquist, Setty and Vestman (2018) solve for a default fund's allocation given a pension system; we vary the system and hold the fund menu fixed, which isolates the same interaction from the other side.

On the decumulation side the rule that reverses our reversal is the actuarial or amortisation rule of Milevsky and Huang (2011). Its interaction with pension design appears not to have been measured: the withdrawal-rule literature conditions on a fixed income floor, and the pension literature conditions on a fixed withdrawal rule.

Our panel is the Jordà–Schularick–Taylor macrohistory database, the same source the replicated study uses, and the same 16 developed markets Dimson, Marsh and Staunton (2002) cover for a comparable period. Every number below is regenerated from that panel by the pipeline described in Section 3; the fuller robustness apparatus, and the twenty-seven extensions not needed for this argument, are in the companion study.

2. Data

2.1 What the panel is

The panel is an annual, country-by-year matrix of five real series — domestic equity, international equity, long-term government bonds, short-term bills and consumer price inflation — for 16 developed markets over 1890–2020. It contains 2,010 usable country-years of domestic equity returns, with a median country history of 125 years and a minimum of 115 years.

All series are *real*: nominal total returns are deflated by each country's own consumer price index in the same year, so that a hyperinflation shows up as a catastrophic real return rather than as a spectacular nominal one. This is not a cosmetic choice. Several of the worst episodes in the sample — Germany in the early 1920s, central Europe in the late 1940s — are invisible in nominal space and dominate the left tail in real space.

The equity series across the panel average 6.8% in arithmetic real terms and 4.3% geometrically, with a mean cross-country standard deviation of 22.2%. The gap between the arithmetic and geometric means — over three percentage points — is itself the story: at this level of volatility, the compounded experience of a lifetime investor is far below the average annual return, and any simulation that draws i.i.d. from the arithmetic mean will overstate what a real investor would have received.

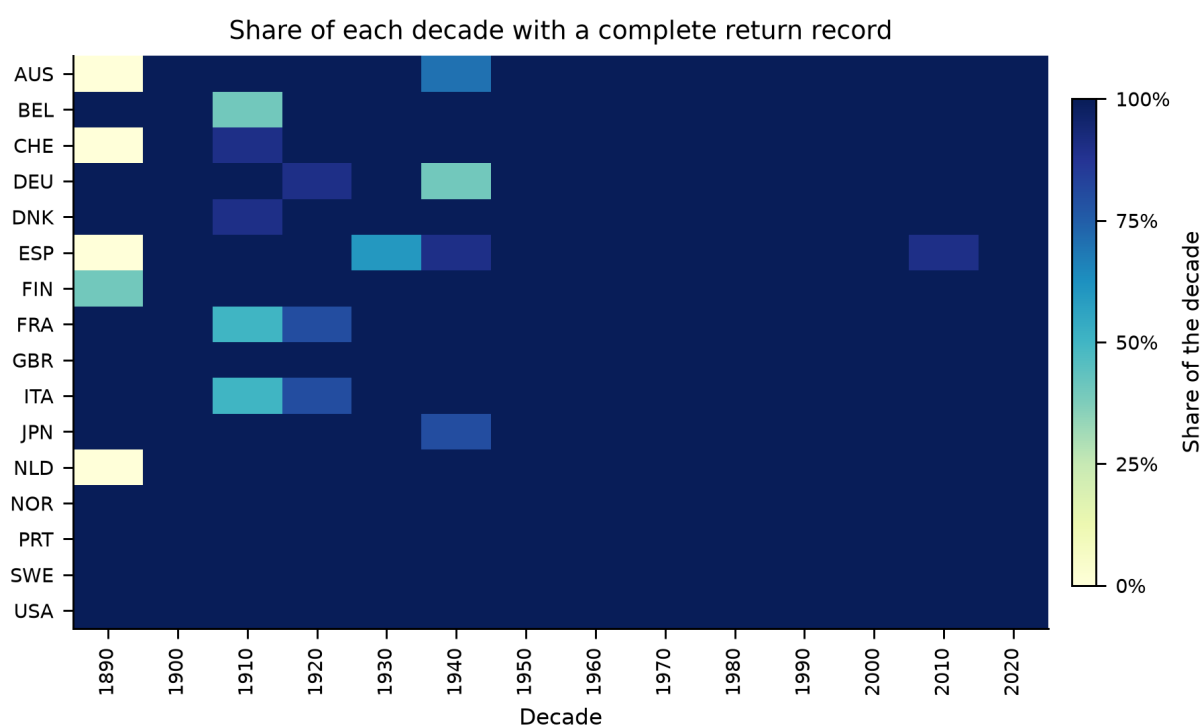


Figure 1.

Share of each decade for which a country has a complete return record. Darker is more complete; the scale is a share so that the final, partial decade compares with the rest. The gaps are not random: they cluster around the two world wars and around the market closures that follow them, which is precisely the period a survivorship-prone sample would drop.

2.2 The country set

The Jordà–Schularick–Taylor database carries complete, independently sourced equity, bond and bill total returns for 16 countries: Australia, Belgium, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom, United States. That set is the panel, and every country-year of every return series in it is a recorded observation.

The wider developed-market universe — IMF advanced economies with an investable domestic equity market, plus Poland — numbers 38. The histories of the other 22 exist in commercial databases such as Global Financial Data and Dimson–Marsh–Staunton, which are proprietary and not redistributable. Working from openly licensed sources, 16 is the extent of the recorded evidence, and Section 2.6.1 reports what the excluded countries do carry.

That breadth is the paper's principal limitation and Section 9.1 develops it. Its most direct consequence is on the international leg, which is a leave-one-out average and therefore spans 15 foreign markets, all advanced economies with long histories. Every statement here about international diversification is a statement about that set.

2.3 Constructing the international leg

An investor in country i holding "international equity" does not hold a global index that includes their own market. For each country-year we therefore build the international return as a leave-one-out average across all other countries with data that year, computed in gross terms and in a common currency before being converted back to the home investor's real terms. Leaving the home market out is what makes the domestic and international legs genuinely distinct assets rather than two overlapping slices of the same one.

Three observations in the raw international series are extreme enough to dominate any lifetime that draws them, all of them artefacts of a market reopening after wartime closure with a collapsed price index. We winsorise the international leg at the 0.1th percentile of its own distribution; the affected observations are listed in full in Table 2 so that the reader can judge the intervention rather than take it on trust.

Table 1. Real domestic equity returns by country

Country	Years	From	To	Mean	Geo. mean	S.d.	Skew	AR(1)
Australia	118	1900	2020	7.8%	6.4%	16.8%	-0.31	-0.07
Belgium	125	1890	2020	5.9%	3.8%	20.7%	-0.00	0.08
Switzerland	120	1900	2020	6.8%	5.1%	18.9%	0.30	0.10
Germany	124	1890	2020	8.7%	4.6%	28.9%	1.08	-0.01
Denmark	130	1890	2020	7.5%	6.1%	17.2%	0.43	0.04
Spain	115	1900	2020	6.2%	4.2%	21.1%	0.77	0.32
Finland	125	1896	2020	8.9%	5.0%	30.0%	1.21	0.18
France	124	1890	2020	3.6%	1.1%	23.3%	0.86	0.15
United Kingdom	131	1890	2020	6.6%	4.9%	18.7%	0.90	-0.05
Italy	124	1890	2020	6.1%	2.6%	26.8%	0.66	-0.04
Japan	129	1890	2020	7.9%	4.2%	25.8%	0.26	0.12
Netherlands	121	1900	2020	7.1%	5.1%	21.2%	0.74	0.03
Norway	131	1890	2020	5.7%	3.7%	20.1%	0.46	0.02
Portugal	131	1890	2020	3.2%	-0.1%	26.0%	0.66	0.25
Sweden	131	1890	2020	8.1%	6.1%	20.3%	0.05	0.13
United States	131	1890	2020	8.4%	6.7%	18.7%	-0.24	0.00

Notes. Arithmetic and geometric means are annual real returns deflated by each country's own consumer price index; AR(1) is the first-order autocorrelation of the annual series. Appendix B repeats this with excess kurtosis.

2.4 Cross-asset structure

The correlation structure the bootstrap must preserve is reported below. Domestic and international equity correlate at 0.43, which is high enough that international diversification is not free and low enough that it is worth having. Bonds and bills correlate at 0.72. Inflation correlates negatively with every asset, most strongly with bills (-0.66), which is the mechanism by which cash loses in real terms over long horizons.

Table 2. Pooled cross-asset correlation matrix of annual real returns

	Dom. equity	Intl. equity	Bonds	Bills	Inflation
Domestic equity	1.000	0.429	0.315	0.234	-0.121
International equity	0.429	1.000	0.110	0.117	-0.042
Bonds	0.315	0.110	1.000	0.716	-0.469
Bills	0.234	0.117	0.716	1.000	-0.662
Inflation	-0.121	-0.042	-0.469	-0.662	1.000

Notes. Pooled across all country-years in the panel. These are the moments the block bootstrap is required to reproduce; Table 4 reports how closely it does so.

Table 3. Every winsorised observation in the international leg

Year	Country	Raw return	After winsorisation
1942	Italy	1120%	301%
1945	Japan	-86%	-70%
1946	France	308%	301%

Notes. All three are markets reopening after wartime closure against a collapsed price index. Left unwinsorised, a single one of these draws would dominate the terminal wealth of any lifetime that contained it.

2.5 Market disruptions and the survivorship question

There are 16 contiguous runs of missing data across the countries of the panel, concentrated in the two world wars. The treatment of these gaps is a substantive modelling decision rather than a data-cleaning one. Interpolating across them would smooth away exactly the episodes that motivate the whole exercise; dropping the countries would reintroduce the survivorship bias the panel exists to avoid.

We do neither. Gaps are preserved as gaps, and the bootstrap is constrained to draw only blocks that lie entirely within a contiguous run of available data for the country in question. A country with a six-year wartime hole contributes blocks on either side of it but never across it. The consequence is that the sampler is *less* able to draw from countries with disrupted histories at long block lengths, which we quantify rather than assume: the run-length admissibility statistics are reported in Section 3.1.

2.6 Auditing the data

The database is used here through a redistributed copy rather than obtained from its compilers, which deserves more scepticism than a citation. This section asks whether the file is genuine, reports one test it does not pass, and describes three bodies of recorded data that sit in the sources and stay out of the model.

2.6.1 Why the panel stops at sixteen

The boundary is set by equity. Four countries outside the panel do carry recorded interest-rate histories: long-term yields and short rates in the macro file for Canada and Ireland, Clio-Infra long yields for Austria and New Zealand. Those are enough to build real bond and bill returns — a bond total return follows from a yield and a duration, $r_t = y_{t-1} + D(y_{t-1} - y_t)$; a bill return is a lagged short rate; and deflating both by that country's own price index gives a real return that was measured rather than modelled.

Table 4. Recorded series that exist and still cannot be used

Country	Series	Source	From	To	Observed years
Austria	bond	Clio-Infra bond yields	1966	2010	45
Canada	bill	Jordà–Schularick–Taylor yields and short rates	1935	2020	86
Canada	bond	Jordà–Schularick–Taylor yields and short rates	1891	2020	130
Ireland	bill	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
Ireland	bond	Jordà–Schularick–Taylor yields and short rates	1923	2020	98
New Zealand	bond	Clio-Infra bond yields	1934	2010	74

Notes. Rebuilt from published rates and deflated by each country's own price index. None of these countries has an equity return series in any source available to us, which is why none is in the panel.

None of it gets them into the panel. A lifecycle investor needs a domestic *equity* return, and the macro file carries none for these four — not even the interpolated variants it supplies elsewhere. Rates alone cannot support a portfolio choice. That is why the panel is sixteen countries and not twenty.

2.6.2 The fourth asset class

The macro file carries a fourth asset the source project measured: **housing total returns**, empirical for all 16 observed countries over 1,805 country-years (1891–2020). It is held out of the headline results, which use the same four-asset set as the paper being replicated, and audited here so the reason is visible rather than assumed. Section 24 of the companion study then puts it into the opportunity set and prices it.

Table 5. Housing total returns: observed, and why they stay out

Country	Years	Mean real	s.d. published	s.d. de-smoothed	Equity s.d.	Autocorr.
Australia	120	6.2%	11.7%	11.7%	16.2%	-0.02
Belgium	119	7.9%	15.2%	21.8%	22.9%	0.34
Switzerland	119	5.7%	6.5%	10.8%	18.9%	0.47
Germany	96	6.6%	8.5%	8.5%	33.5%	-0.13
Denmark	130	8.0%	7.8%	12.1%	17.2%	0.42
Spain	120	5.3%	11.8%	12.2%	20.8%	0.03
Finland	99	9.4%	15.2%	24.0%	30.0%	0.47
France	130	6.3%	10.3%	21.5%	23.1%	0.63
United Kingdom	118	5.4%	9.0%	12.5%	18.7%	0.33
Italy	86	4.7%	9.3%	9.3%	26.6%	-0.05
Japan	75	6.6%	8.1%	12.0%	25.8%	0.46
Netherlands	130	7.4%	9.0%	14.9%	21.2%	0.46
Norway	130	7.8%	8.6%	10.3%	20.1%	0.17
Portugal	73	6.6%	8.5%	11.5%	26.0%	0.33
Sweden	130	8.1%	8.9%	12.8%	20.3%	0.35
United States	130	6.2%	7.9%	7.9%	18.7%	-0.17

Notes. Real annual total returns on the national housing stock, from the same source as the equity series. De-smoothing is first-order Geltner using each country's own lag-one autocorrelation.

The comparison is the one the source project is known for. Median real housing returns are **6.6%** against **6.9%** for the same countries' equity — indistinguishable — at a published standard deviation of 8.9% versus 21.0% — a ratio of 0.43. Equity-like returns at that volatility would look dominant in any mean-variance comparison in this paper, which is precisely why the series deserves scrutiny before adoption.

A house price index is built from appraisals and sparse transactions, and that smooths it. The median lag-one autocorrelation of housing returns is 0.34 against 0.06 for equity, and housing is the more autocorrelated series in 12 of 16 countries. Undoing that to first order raises the median standard deviation from 8.9% to 12.0%: most of the apparent free lunch is a measurement artefact.

Even de-smoothed the series is not investable as written. It is an unlevered, untaxed, frictionless total return on an entire national housing stock, with no transaction costs, no vacancy, no maintenance and none of the single-property concentration a real household bears. Adopting it as a fourth sleeve at face value would therefore overstate what a household can buy. Section 24 of the companion study adds it anyway, but only after de-smoothing it and charging an explicit annual holding cost — and it is the size of that cost, not the raw return, that decides the answer.

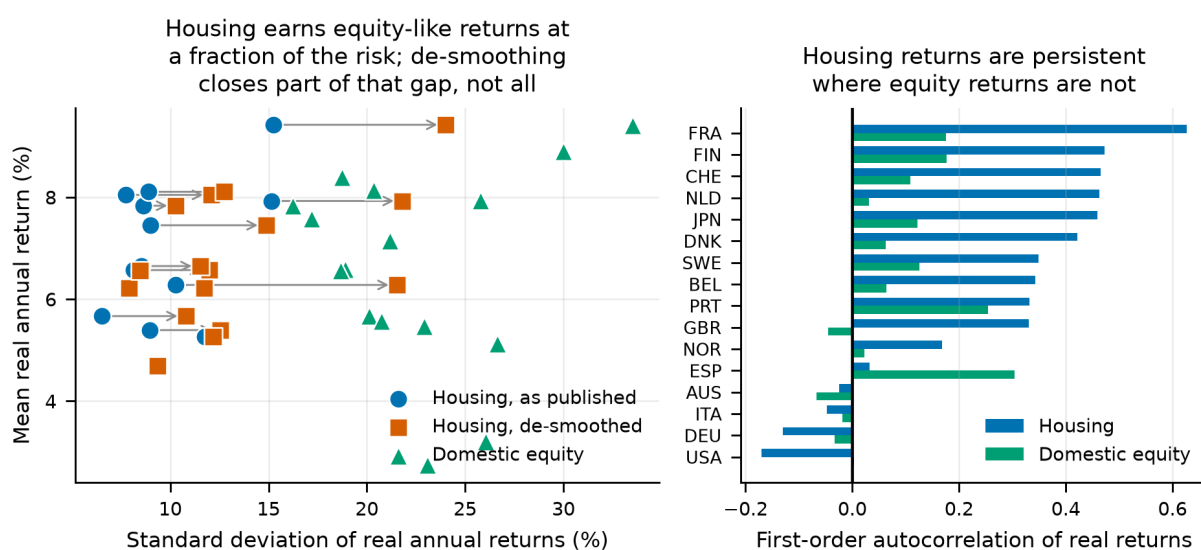


Figure 2.

Left: risk and return by country for housing as published, for housing with its own first-order smoothing undone (arrows), and for that country's domestic equity. Right: the lag-one autocorrelation that does the work. De-smoothing closes part of the volatility gap and not all of it, which is why the holding cost of Section 24 of the companion study has to carry the rest of the argument.

2.6.3 The series that bears on our income model

The same file carries a nominal wage index for all 18 of its countries, including the two with no return series. Deflated by each country's own consumer prices it measures economy-wide **real wage growth** over 2,269 country-years (1891–2020). The median country compounded real wages at **1.41% a year**, from -0.02% (Belgium) to 3.16% (Finland) — which over the 37-year career we simulate compounds to 1.68×

Table 6. Measured real wage growth, and what our income profile assumes

Country	Years	Real wage growth p.a.	Excl. war years	s.d.	Compounded over a career
Australia	130	1.43%	1.61%	4.0%	1.72
Belgium	124	-0.02%	1.20%	17.3%	0.99
Canada	130	1.36%	1.27%	2.5%	1.67
Switzerland	130	1.62%	1.75%	3.8%	1.84
Germany	128	0.88%	1.03%	7.9%	1.39
Denmark	130	2.01%	2.05%	6.1%	2.13
Spain	122	1.98%	2.68%	10.3%	2.11
Finland	130	3.16%	3.81%	15.5%	3.26

Country	Years	Real wage growth p.a.	Excl. war years	s.d.	Compounded over a career
France	130	1.11%	1.85%	5.2%	1.52
United Kingdom	130	1.29%	1.41%	4.6%	1.63
Ireland	77	2.83%	2.75%	3.2%	2.89
Italy	128	0.98%	2.26%	10.7%	1.45
Japan	130	2.37%	2.66%	13.8%	2.43
Netherlands	130	1.12%	1.18%	4.7%	1.53
Norway	130	1.83%	1.79%	3.8%	1.99
Portugal	130	0.74%	1.03%	9.9%	1.32
Sweden	130	2.43%	2.56%	4.3%	2.49
United States	130	1.40%	1.31%	2.5%	1.69

Notes. Geometric means, because they are what compound across a career. Our deterministic profile implies 1.18% a year over the same span.

The war years carry more of that spread than the economics does, which is why the table reports the series both ways. The two largest observations in the panel are Belgium 1919 (+151%) and Finland 1918 (-66%): a wage index spanning occupation, rationing, suppressed prices and post-war repricing is measuring those at least as much as it is measuring wages. Dropping 1914–1919 and 1939–1946 lifts the median to 1.77% (+0.36 percentage points) and moves the lowest country, Belgium, from -0.02% to 1.20% — from an implausible claim about a century of that country's wages to an unremarkable one. We keep them in the headline number, because a worker alive then lived through them, and note that excluding them only widens the gap we are about to describe.

Our income profile has no term for it. Section 3.3 sets real labour income as a deterministic hump peaking at age 50 at 1.76× starting income and ending the career at 1.54× — an average of 1.18% a year. That is an *age* effect: the progression a worker earns by getting older. Economy-wide wage growth is a different quantity, lifting the whole distribution regardless of age, and in the Cocco–Gomes–Maenhout estimation our profile is taken from the two are separated by construction and are therefore additive. A worker facing both would see roughly 2.61% a year.

That caveat decides the size of the gap rather than its existence: whether the components add depends on how the source profile was estimated, and one fitted to a panel that still carried time effects would already absorb part of the growth. What is not in doubt is that the quantity is measurable, that eighteen countries measure it, and that nothing in our pipeline reads it.

The direction is the one that matters for reading this paper. Understating income growth understates human capital throughout the career, and human capital is the bond-like asset in the standard lifecycle argument — so having less of it *weakens* the case for holding equity when young. The bias runs against our conclusion rather than toward it, as most of the known biases here do. The effect on the savings analysis of Sections 26 of the companion study and 27 of the companion study is genuinely ambiguous, because faster income growth raises both what a given savings rate accumulates and the consumption it must replace, and this audit does not resolve it. Re-estimating the income process is a modelling change rather than a data one, so we record it as a quantified limitation rather than apply it silently.

2.6.4 Is the source we kept genuine?

The workbook was obtained from a redistributed copy rather than downloaded from the compilers, so it is audited rather than assumed. Two tests, in opposite directions.

Table 7. The workbook against independently known annual returns

Observation	This workbook	Independently known	Difference	Passes
US equities in the worst year of the slump	-0.4030	-0.4300	0.0270	yes
US equities in the global financial crisis	-0.3875	-0.3700	-0.0175	yes
US equities in the 1974 bear market	-0.2544	-0.2600	0.0056	yes
US equities rebounding in 1933	0.5264	0.5400	-0.0136	yes
US equities in 2013	0.2956	0.3200	-0.0244	yes

Notes. Nominal equity total returns for years whose magnitude is not in dispute. A reconstructed or synthetic file would not reproduce both directions of the 1931-33 swing and the 2008 drawdown.

All 5 of 5 land within tolerance. The second test runs the other way. Equity total return should satisfy the accounting identity $eq_tr = (1 + capital\ gain)(1 + dividend\ yield) - 1$; across 2,163 observations it fails by more than one part in a million 38% of the time. **That failure rate is evidence of authenticity.** The components in the real database are sometimes spliced from different underlying indices, so they do not always reconcile; a file that satisfied the identity everywhere to machine precision would be one whose components had been back-solved from its totals.

2.6.5 One finding that does not pass

The last five years of every equity series are smoother than that country's own history. **All 16 of 16 countries** show a lower standard deviation over 2016–2020 than over 1950–2015, with a median ratio of 0.54. A five-year standard deviation is far too noisy to read one country at a time, which is why this is a sign test: under a fair-coin null the probability of all 16 pointing the same way is **3.1e-05**. Spot-checking the United States sharpens it — the workbook records +14.2% for 2018, a year in which every broad US index fell, and +8.2% for 2019 against roughly +31% for the S&P 500.

The likely explanation is that the redistributed copy was extended past the compilers' own end date by someone else. **We therefore treat 2016–2020 as unverified.** Those five years are under four percent of the panel's country-years and cannot move a 68-year lifecycle result materially, but the finding is recorded rather than left for a reader to discover.

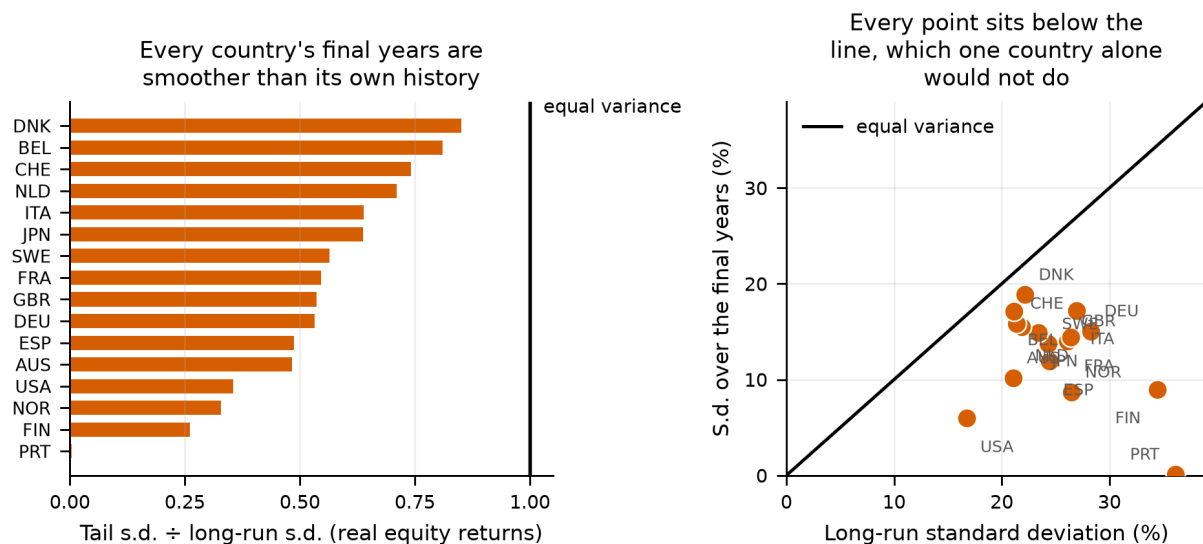


Figure 3.

The variance test, read two ways. Left: each country's standard deviation over the final five years as a ratio of its own 1950–2015 standard deviation. Right: the two standard deviations plotted against each other, with the 45-degree line. One country below the line would be unremarkable; every country below it is a property of the file rather than of the markets.

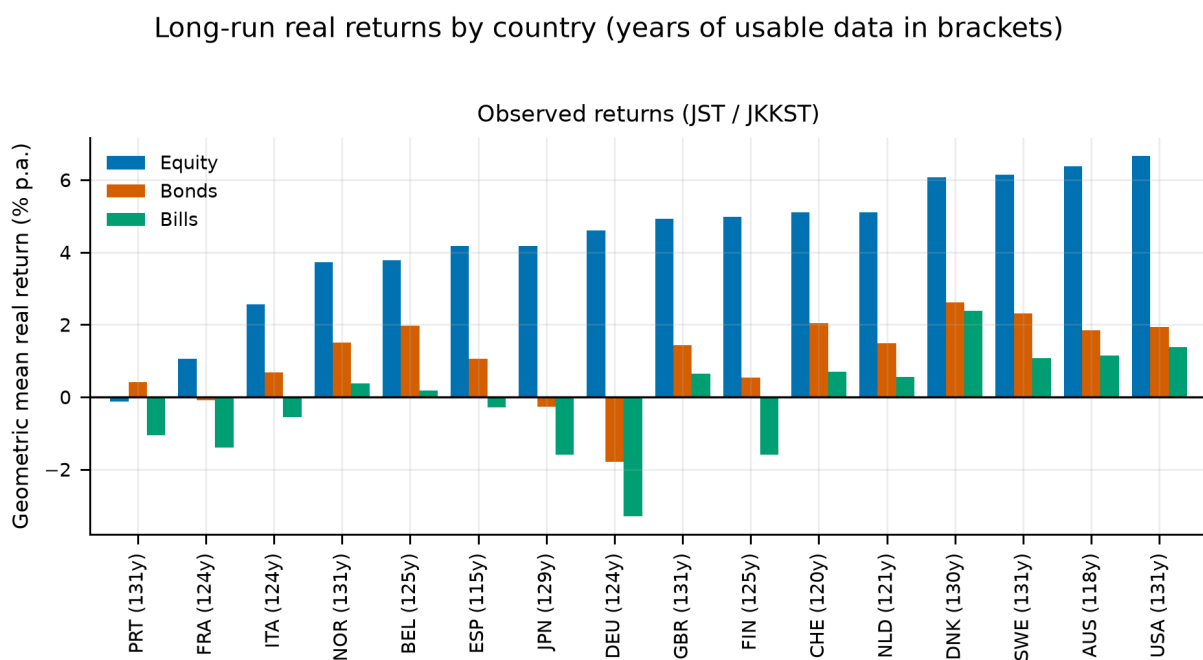


Figure 4.

Distribution of annual real returns by country and asset class. The left tails are the point of the panel: several countries record single-year real equity losses beyond –60%, and these are not outliers to be trimmed but the events a lifecycle investor is exposed to.

3. Methodology

3.1 The cross-country stationary block bootstrap

A simulated lifetime is 68 years long and is assembled from contiguous blocks of history. The sampler is a stationary block bootstrap in the sense of Politis and Romano (1994): block lengths are drawn from a geometric distribution with mean 10 years, truncated to [1, 40], and blocks are laid end to end until the horizon is filled.

Two features distinguish it from a textbook block bootstrap and both matter for the result.

- **Blocks are calendar-joint.** A block is a (country, start-year, length) triple, and the same triple is applied to all five series simultaneously. Domestic equity, international equity, bonds, bills and inflation for a given simulated year therefore come from the same real country in the same real year. This is what preserves the cross-asset correlation matrix of Table 2 without any of it being imposed parametrically.
- **Blocks respect data gaps.** Admissibility is enforced through pre-computed run-lengths: for each (country, year) the sampler knows how many consecutive years of complete data follow, and only draws a block that fits inside a run. A wartime hole is never bridged.

Across the production run the sampler drew 30,848 blocks with a mean realised length of 8.82 years against a target of 10, a median of 6, a ninetieth percentile of 20 and a maximum of 40. The realised mean falls short of the target for a mechanical reason worth stating: truncation at the maximum and the gap-admissibility constraint both bite from above, so the effective block length is always slightly below the nominal one.

The country for a lifetime is drawn once (*per_lifetime*) with probability proportional to the length of that country's usable history (*history*). Both choices are varied in Appendix C: redrawing the country at every block, and weighting countries uniformly, both leave the ranking unchanged. The weighting choice has little room to bite here, because every country carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. The choice would matter on a panel with short histories, where a country covering only one era could otherwise speak as loudly as one covering a century and a half.

Table 8. Does the bootstrap reproduce the panel it samples from?

Series	Bootstrap mean	Panel mean	Gap (bp)	Bootstrap s.d.	Panel s.d.	s.d. ratio	Bootstrap kurtosis
Domestic equity	6.86%	6.77%	8.3	22.87%	22.50%	1.016	2.9
International equity	8.52%	8.33%	19.0	22.27%	21.69%	1.027	42.5
Bonds	2.02%	1.98%	3.9	11.51%	11.81%	0.974	6.0
Bills	0.58%	0.44%	14.0	7.37%	7.99%	0.922	24.0
Inflation	5.23%	5.97%	-74.1	18.76%	37.66%	0.498	1282.5

Notes. Means agree to within twenty basis points on every series and standard deviations to within eight percent. The inflation row has an excess kurtosis in the thousands: that is the hyperinflation episodes surviving into the simulated paths, which is the intended behaviour rather than a defect.

The correlation matrix is reproduced to a maximum absolute deviation of 0.031 across the four asset-to-asset pairs that drive portfolio behaviour. The largest deviation anywhere in the matrix is 0.068 and it sits on the inflation row, where the excess kurtosis reported above makes the sample correlation itself unstable. We report both figures rather than the flattering one.

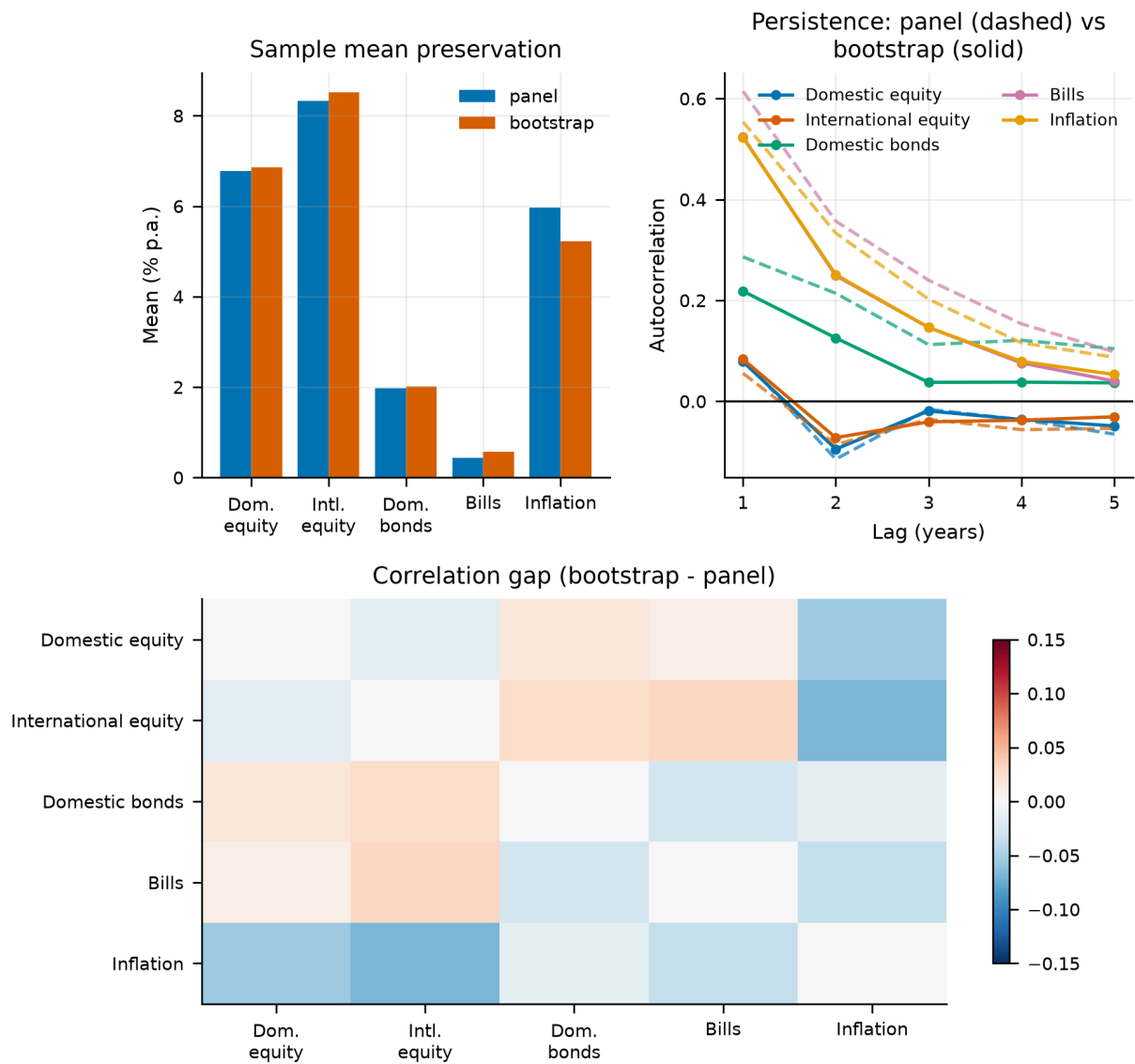


Figure 5. Bootstrap diagnostics. Simulated marginal distributions against the panel they are drawn from, the preservation of the cross-asset correlation structure, and the within-path autocorrelation that block sampling is designed to retain.

Block length and long-horizon dispersion

Block length is the single most consequential tuning parameter in the design, because it controls how much multi-decade persistence survives into a simulated lifetime. The naive expectation is that longer blocks produce more dispersion in long-horizon outcomes. The data say the opposite, and the reason is instructive.

Table 9. Sensitivity of 68-year outcomes to the mean block length

Mean block (years)	Mean annualised	S.d. annualised	5th pct	95th pct	Within-path AR(1)
1	4.41%	3.31%	-1.52%	9.49%	-0.014
5	4.49%	3.27%	-1.41%	9.25%	0.064
10	4.43%	3.20%	-1.53%	8.89%	0.079
20	4.37%	3.14%	-1.69%	8.68%	0.088

Notes. Longer blocks raise the within-path autocorrelation, as intended, but slightly *reduce* the dispersion of annualised 68-year outcomes. Over a horizon this long the historical series mean-reverts, so preserving more of its internal structure preserves that mean reversion too.

Within-path autocorrelation rises monotonically from -0.014 at one-year blocks to 0.088 at twenty-year blocks, confirming that the sampler is doing what block sampling is for. But the standard deviation of annualised 68-year returns *falls* slightly over the same range. This is a long-horizon mean-reversion finding, not a bug: a lifetime that inherits real historical sequences inherits their tendency to revert, whereas one assembled from independent annual draws does not. It is also a caution against reading the block length as a simple dial for "more risk".

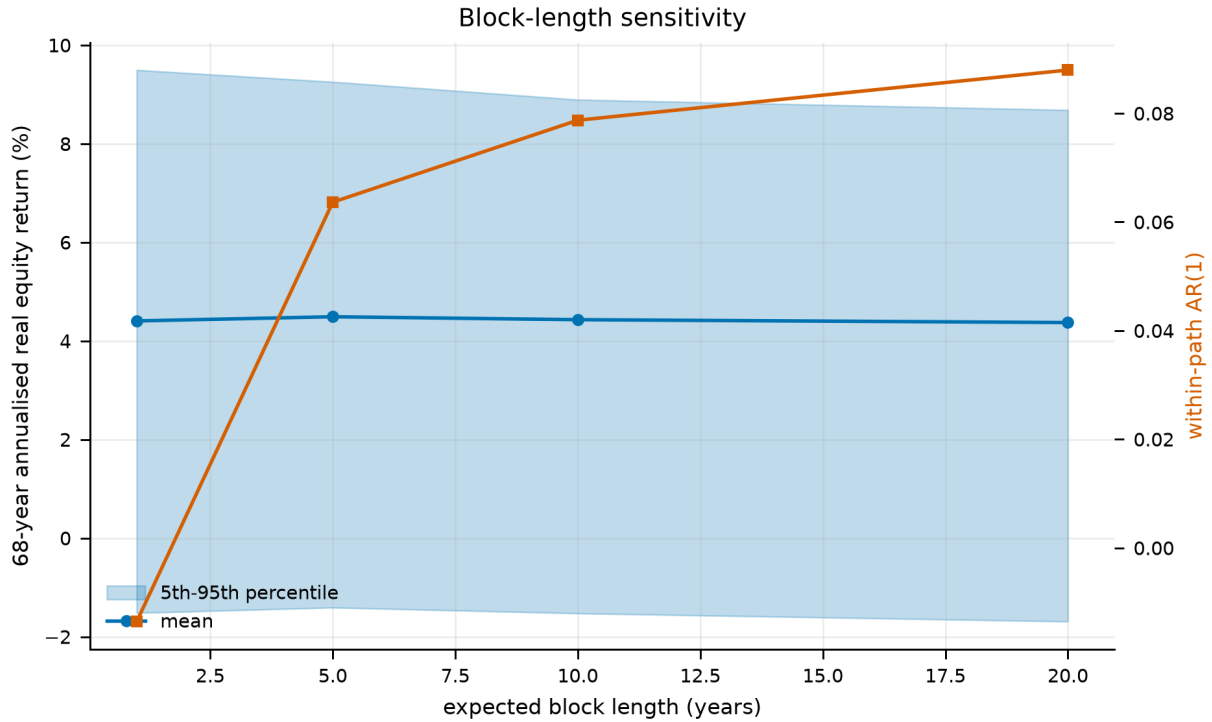


Figure 6.

Block-length sensitivity. Long-horizon dispersion is close to flat in the mean block length even as within-path persistence rises steadily, which is the signature of mean reversion in the underlying series.

3.2 The lifecycle model

A simulated investor begins work at age 25, retires at 63 and dies at 93, giving a 68-year horizon of which 38 are working years. They save a fixed fraction $s = 10\%$ of labour income, rebalance annually to the strategy's target weights, and on retirement switch to a withdrawal rule.

Wealth evolves as

$$W_{h+1} = (W_h + c_h - x_h) \cdot (1 + r_h^p), \quad r_h^p = \sum_a w_{a,h} r_{a,h} \quad (1)$$

where c_h is the contribution in working years and zero afterwards, x_h is the withdrawal in retirement and zero before, $w_{a,h}$ is the strategy's weight on asset a at age h , and all returns are real. Consumption is labour income net of saving while working, and the sum of the social-security benefit and the portfolio withdrawal in retirement:

$$C_h = (1 - s) Y_h \text{ (working)} \quad C_h = B + x_h \text{ (retired)} \quad (2)$$

Labour income follows a deterministic hump-shaped profile ($b_1 = 0.045$, $b_2 = -0.0009$ in age and age squared) multiplied by permanent and transitory shocks with standard deviations 0.10 and 0.25. The permanent shock is a cumulative sum, so income dispersion widens over a career; the transitory shock is i.i.d. The profile is an age effect and carries no economy-wide wage growth; Section 2.6.3 measures what that leaves out and which way it

biases the result.

The social-security benefit uses the US primary-insurance-amount bend-point schedule rather than a flat replacement rate, applying 90%, 32% and 15% to successive tranches of career-average earnings. This is not decoration. A flat replacement rate scales with the outcome and therefore insures nothing; a progressive schedule supplies a genuine real consumption floor, which is exactly what determines how much the left tail of a risky strategy actually costs.

Table 10. The six benchmark strategies, as constant weights

Strategy	Domestic eq.	Intl. eq.	Bonds	Bills
50/50 domestic/international equity	50%	50%	—	—
100% domestic equity	100%	—	—	—
100% international equity	—	100%	—	—
60/40 domestic equity/bonds	60%	—	40%	—
100% bills (cash)	—	—	—	100%
Target-date fund (glide path)	see below	see below	see below	see below

Notes. Held constant at every age and rebalanced annually, except the target-date fund.

The target-date fund is the one benchmark whose weights move. Its equity share is piecewise-linear in age through the knots below, interpolated between them; the equity sleeve is split 60/40 domestic/international at every age and the fixed-income sleeve 70/30 bonds/bills. The shape is the industry standard — flat and growth-heavy while young, declining through the decade before retirement, and still declining gently after it.

Table 11. The target-date fund's glide path, at its knots

Age	25	40	55	63 (retire)	75	93
Equity share	90%	90%	65%	50%	35%	30%

Notes. Linear between knots. At age 63 the fund therefore holds 30% domestic equity, 20% international, 35% bonds and 15% bills.

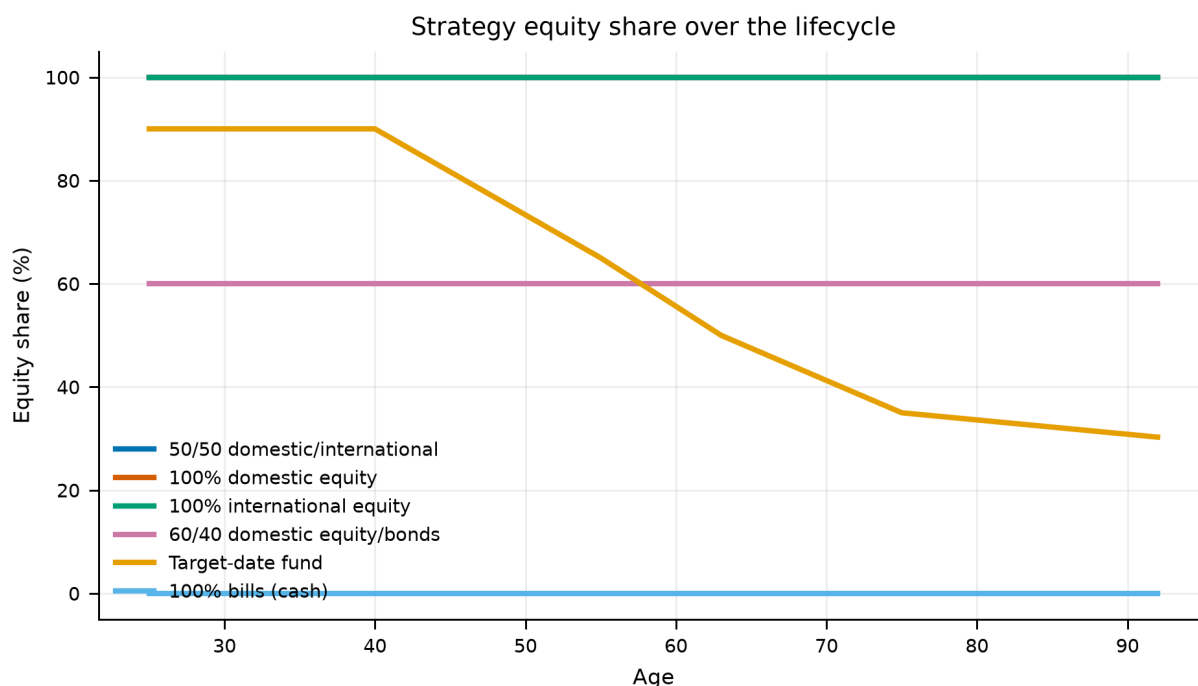


Figure 7.

The six benchmark allocation strategies as equity share by age. The target-date fund is the industry-standard declining glide path; the comparison strategies are held at fixed weights.

3.3 Preferences and the certainty equivalent

Strategies are ranked by certainty-equivalent consumption: the constant, riskless consumption stream that would leave the investor indifferent to the risky one their strategy produces. Under constant relative risk aversion with discount factor β and curvature γ , lifetime utility over the evaluation window is

$$V = \sum_h \beta^h u(C_h) + \beta^H \theta \cdot u(\kappa + W_H), \quad u(c) = c^{1-\gamma} / (1 - \gamma) \quad (3)$$

and the certainty equivalent inverts that utility back into consumption units. We report risk aversions of $\{2, 5, 10\}$ with $\gamma = 5$ as the baseline, $\beta = 0.96$, and a bequest weight $\theta = 2$.

The bequest term uses the shifted-power specification of De Nardi (2004) with $\kappa = 1$. The shift is not cosmetic. Without it, any path that dies with exactly zero wealth contributes $u(0) = -\infty$ at $\gamma \geq 1$, one such path in a hundred thousand collapses the entire certainty equivalent, and the ranking becomes a statement about which strategy avoids exact zeros rather than about which delivers better consumption. This is a genuine trap in lifecycle work and we flag it because we walked into it.

We also report Epstein–Zin preferences in their ex-ante, early-resolution form, which separates risk aversion from the elasticity of intertemporal substitution, with ψ taking the values $\{0.5, 1.5\}$. Under CRRA these two are locked together at $\psi = 1/\gamma$, and a reader is entitled to ask whether the result is being driven by the intertemporal channel rather than the risk channel. Section 6.2 of the companion study shows it is not.

One further specification choice deserves emphasis. Utility is evaluated over the **retirement** window rather than the whole lifetime for the allocation comparisons. The reason is mechanical: with a fixed savings rate, working-life consumption is *identical* across allocation strategies by construction, so including it adds a large constant to every strategy's utility and compresses the differences that the exercise is about. Where a policy *does* change working-life consumption — the retirement timing and savings-rate studies of Sections 11 of the companion study to 28 of the companion study — we switch to the whole-lifetime window and say so explicitly.

3.4 The comparison discipline

Most of the extensions in this paper compare policies that differ in more than one way at once, and the arithmetic of a certainty equivalent will happily credit the wrong difference. Four rules are applied throughout.

- **Matched baselines.** A rule that retires people later is scored against a fixed date matched on the same *mean* retirement age, not against the original age 63. A savings rule that drifts to saving more is scored against a constant rate interpolated at its own realised career average. Without this, a policy is rewarded for working longer or saving more rather than for being smarter.
- **Common random numbers.** Every sweep and every optimiser reuses the same bootstrap draws and the same income shocks, so a difference between two settings is a policy effect and not a sampling artefact. This also makes the objective a deterministic function of the policy, which is what licenses coordinate ascent.
- **Deviation profiles.** Before any structure in a solved schedule is described in words, each element is reset to a neutral reference and the cost measured in basis points. A solved glide path can look highly structured while most of its structure is worth nothing; the profile separates the two.
- **Grid-edge checks.** An optimum sitting on the boundary of its own search grid is a truncation, not an optimum. Every such case is flagged in the text and, where it mattered, the grid was widened and the analysis rerun.

These are not stylistic preferences. Section 28 of the companion study reports a result that fell by roughly sixty percent when a matched baseline replaced an unmatched one, Section 19 of the companion study reports apparent glide-path structure that the deviation profile dissolved, and Section 27 of the companion study reports a functional-form ranking that reversed its interpretation once the grid-edge check was applied.

3.5 Implementation and verification

The model is implemented in Python with NumPy and pandas. Portfolio returns for a whole cohort are formed as a batched matrix product and the wealth recursion runs vectorised across paths, which is what makes a hundred thousand lifetimes per strategy — and the tens of thousands of policy evaluations the optimisers require — tractable on a single core.

Correctness is enforced by 1379 automated tests. The load-bearing ones are equivalence tests: the path-dependent engine used for flexible retirement and conditional saving must reproduce the fixed-date engine *bit for bit* when given a fixed date and a constant rate, and the batched glide-path evaluator must reproduce the reference simulator to stated floating-point tolerance. Every extension in this paper is built on a simulator that is asserted equal to the one that produced the baseline.

Appendix D sets out how the computation is organised, what the test suite establishes, and the discipline by which the prose in this paper is held to the tables it describes.

4. Baseline Results

All results in this section use 100,000 simulated lifetimes per strategy, drawn from the same bootstrap sample so that strategies face identical market histories. Certainty equivalents are over the retirement window, for the reason given in Section 3.3.

4.1 The headline comparison

Table 12. Headline lifecycle outcomes by strategy

Strategy	CEC $\gamma=2$	CEC $\gamma=5$	CEC $\gamma=10$	P(ruin)	Median wealth at 63	Median cons.	5th pct cons.	Median bequest
50/50 domestic/international equity	1.406	1.048	0.756	11.7%	21.2	1.483	0.707	44.7
100% domestic equity	1.183	0.882	0.668	25.4%	16.1	1.241	0.579	16.7
100% international equity	1.521	1.108	0.782	9.0%	24.4	1.613	0.755	64.2
60/40 domestic equity/bonds	1.105	0.871	0.671	24.2%	13.3	1.149	0.585	10.2
Target-date fund (glide path)	1.214	0.943	0.704	22.2%	16.4	1.273	0.630	9.4
100% bills (cash)	0.847	0.739	0.618	56.4%	6.4	0.856	0.521	0.0

Notes. Consumption and wealth are expressed as multiples of the investor's real income at age 25. P(ruin) is the probability of exhausting the portfolio with retirement years still to fund. 100,000 paths per strategy on common bootstrap draws.

The ordering is unambiguous at every risk aversion tested. At $\gamma = 5$ the 50/50 all-equity portfolio delivers a certainty equivalent of 1.048 against 0.943 for the target-date fund, an advantage of 11.2%. Against the 60/40 portfolio the advantage is 20.3%, and against bills 41.8%.

What makes the result more than a restatement of the equity premium is the tail behaviour. The all-equity portfolio has a *lower* ruin probability (11.7%) than the target-date fund (22.2%), a higher fifth-percentile retirement consumption (0.707 against 0.630), and a lower probability of falling below a seventy-percent replacement target (13.6% against 22.6%). The conservative portfolio is not buying downside protection on this panel. It is paying for the appearance of it.

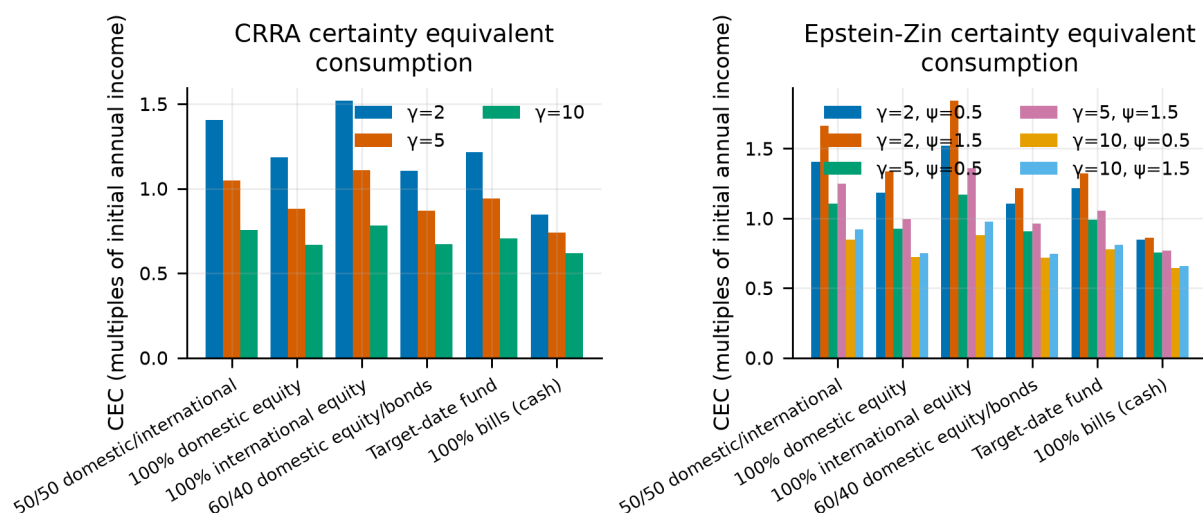


Figure 8.

Certainty-equivalent consumption by strategy and risk aversion. The ranking is stable in γ ; the gap narrows as risk aversion rises but does not close within the range tested.

4.2 The mechanism is international, not equity

Domestic equity alone is *not* the winning strategy. Its certainty equivalent at $\gamma = 5$ is 0.882, below the 50/50 portfolio's 1.048, and its ruin probability is 25.4% — nearly double the diversified portfolio's. Pure international equity does better still (1.108), which is a consequence of the leave-one-out construction rather than a recommendation: an investor holding "international" equity in this model holds a 15-country average, and no single real investor has that opportunity set without also holding their own market.

This is the one place the replication does not reproduce ACO's headline. Their recommended portfolio is an even 50/50 split, held for life; on this panel that split is beaten by the international leg alone. We read the difference as a property of a 16-country panel rather than a correction to them. The domestic leg here is a single draw from a set that includes several century-long underperformers, while the international leg averages away exactly that risk; with thirty-nine countries and a tradeable international index the two legs are far closer in character, and the case for holding both is correspondingly stronger. Nothing measurable here contradicts the 50/50 recommendation — the narrower claim this panel supports is that a diversified sleeve dominates a concentrated one.

The practical reading is that the case for equities here is a case for *diversified* equities. A single national market can and did deliver multi-decade real losses; the cross-section of national markets did not do so simultaneously. This is a claim about the covariance structure of the panel, and it is the reason the exercise needs an international sample rather than a longer US one.

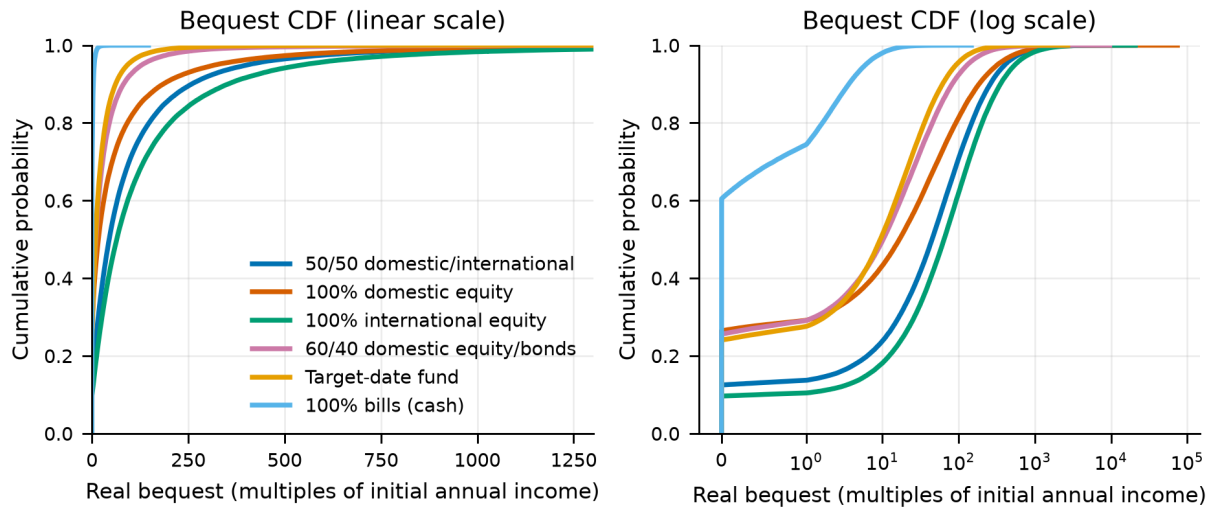


Figure 9.

Cumulative distribution of wealth at retirement. The all-equity distributions stochastically dominate the conservative ones over almost the entire range, including the left tail where the conservative case is usually made.

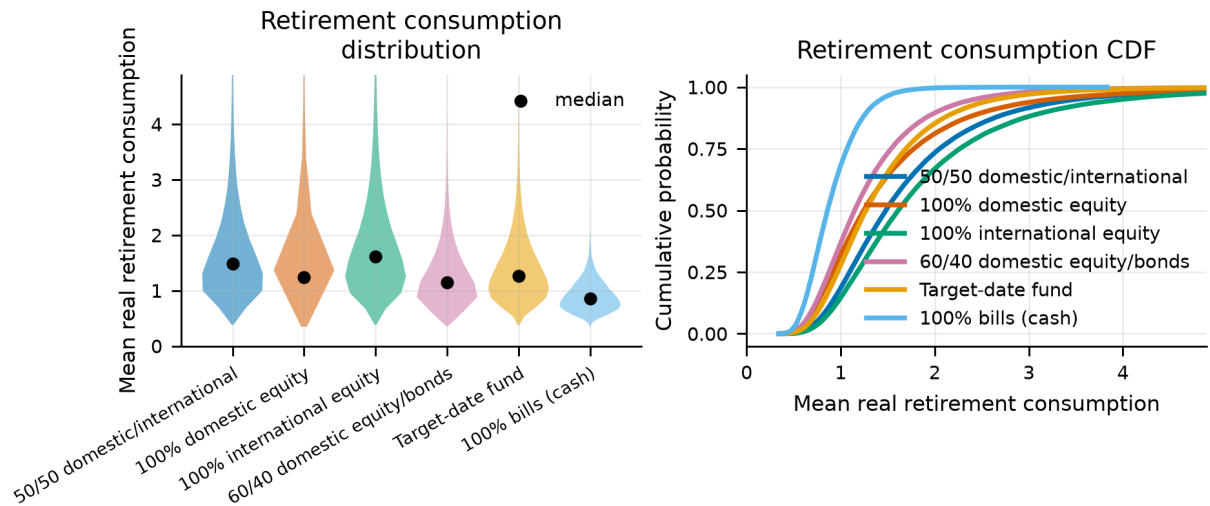


Figure 10.

Distribution of average real retirement consumption by strategy. The vertical line marks the seventy-percent replacement target used in the shortfall statistics.

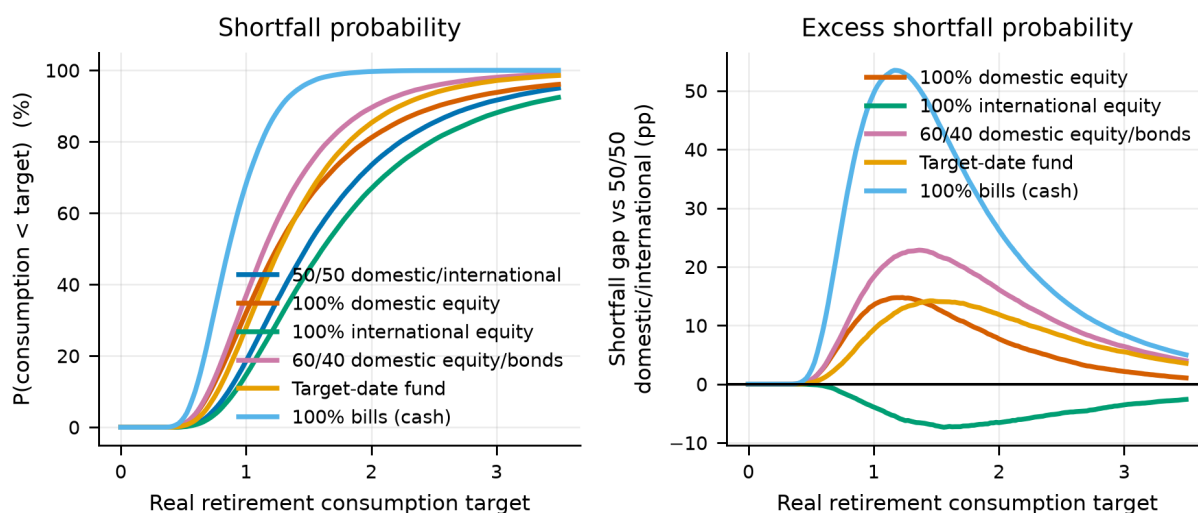
4.3 Distributional dominance

A certainty equivalent is a single scalar and can hide a strategy that wins on average while losing where it matters. We therefore test the all-equity portfolio against each rival across 21 separate distributional criteria — percentiles of wealth at retirement, of retirement consumption and of bequest, plus ruin and shortfall probabilities — counting a tie as neither a win nor a loss.

Table 13. Distributional dominance of the 50/50 all-equity portfolio

Beaten strategy	Criteria	Won	Tied	Lost	Strict dominance
Target-date fund (glide path)	21	20	1	0	yes
60/40 domestic equity/bonds	21	20	1	0	yes
100% domestic equity	21	20	1	0	yes
100% bills (cash)	21	20	1	0	yes

Notes. A tie is counted separately rather than as a loss, since counting it either way would misstate the comparison. The single tie in each row is the zero-bequest floor, which several strategies reach.

**Figure 11.**

Shortfall probabilities against a fixed consumption target. Measuring each strategy against a common target rather than against its own median is essential: a strategy-specific target rewards strategies with low medians.

4.4 Sustainable withdrawal rates

The four-percent rule is the most widely used number in retirement planning. It comes from US historical sequences and it does not survive this panel.

Table 14. Sustainable withdrawal rates on the international panel

Strategy	Sustainable rate at 5% ruin	Ruin probability at 4%
100% international equity	3.48%	9.1%
50/50 domestic/international equity	3.17%	11.7%
Target-date fund (glide path)	2.10%	22.0%
60/40 domestic equity/bonds	1.69%	24.0%
100% domestic equity	1.57%	25.2%
100% bills (cash)	1.07%	56.0%

Notes. The sustainable rate is the highest initial real withdrawal rate, inflation-adjusted thereafter, that leaves at most a five-percent probability of exhausting the portfolio over a 30-year retirement.

At a five-percent ruin tolerance the sustainable rate is 3.17% for the all-equity portfolio and 2.10% for the target-date fund. Withdrawing four percent produces a ruin probability of 11.7% and 22.0% respectively. Two things follow. First, the ordering of strategies is the same here as everywhere else in the paper — the equity portfolio sustains a higher rate, not a lower one. Second, the level is a serious warning: on a sample that includes the twentieth century as most countries lived it, four percent is roughly a one-in-seven proposition even on the best strategy tested.

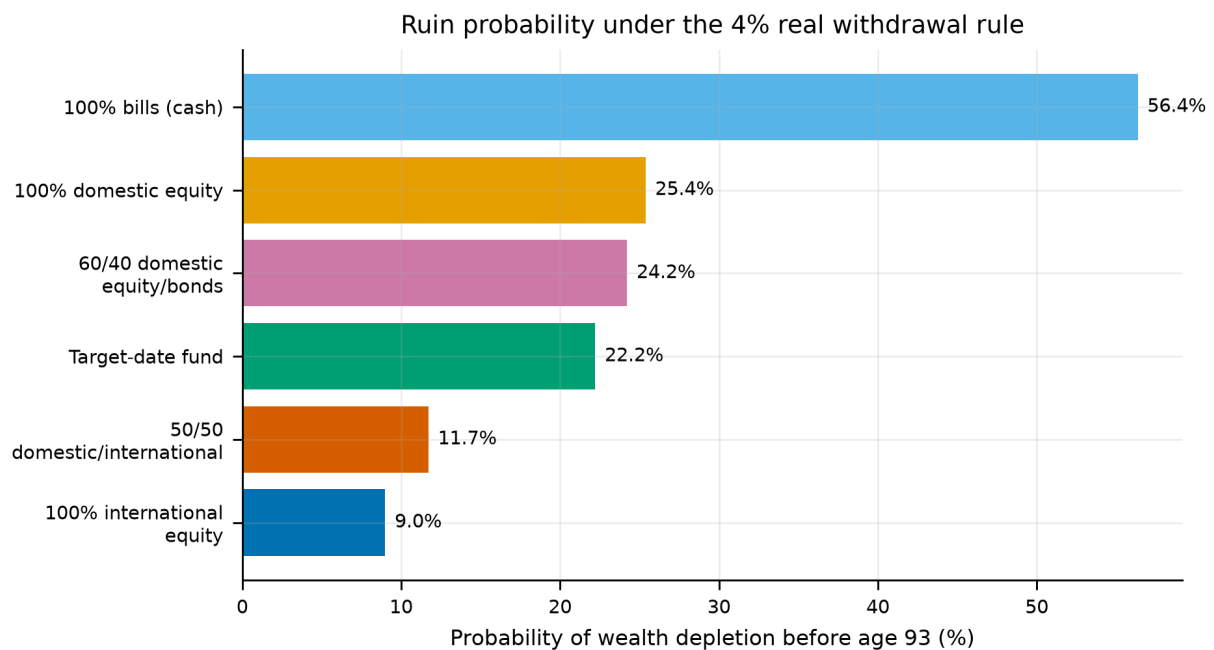


Figure 12.

Probability of portfolio exhaustion as a function of the initial withdrawal rate. The horizontal line is the five-percent tolerance used to define the sustainable rate.

4.5 How the countries are drawn

Two choices in the sampler decide how the cross-section enters a simulated life, and neither is obviously right. A lifetime's domestic country can be drawn once and held, or redrawn at every block; and countries can be weighted by the length of their recorded history or treated as equally likely. Appendix C reports both variants in full. Neither reverses a ranking.

The weighting choice has little room to bite, because every country in the panel carries between 115 and 131 usable years — history weighting and uniform weighting assign nearly the same probabilities. It would matter on a panel containing short histories, where a country covering a single era could otherwise speak as loudly as one covering a century and a half.

Redrawing the country at every block is the more consequential variant, and it is the design the original study uses. It implicitly assumes an investor can be resident in a different country every decade, which is not a description of anybody, but it pools the cross-section more aggressively and so gives the international mechanism its most favourable reading. Holding the country fixed for a lifetime — the specification behind every headline number here — is the more conservative of the two.

Wealth trajectories: median with 25-75 and 5-95 percentile bands

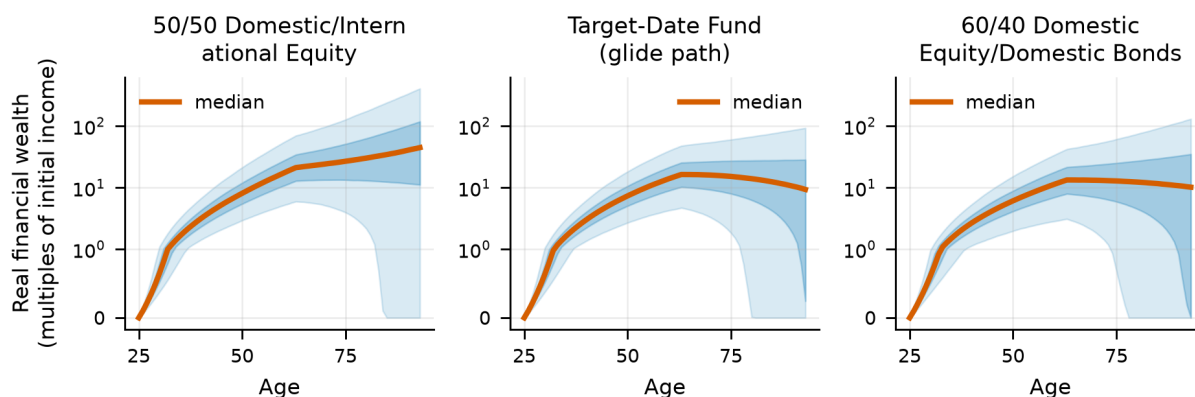


Figure 13.

Wealth trajectories by percentile for the all-equity portfolio and the target-date fund. The fan makes visible what the scalar comparison cannot: the conservative path is not narrower at the bottom, it is lower throughout.

5. One Country's Pension Behind Sixteen Countries' Returns

Every result to this point pays the same public pension in all 16 markets: the US primary-insurance-amount schedule, progressive in career earnings and paid in full whatever else the retiree owns. It is the schedule of exactly one country in the panel, and the panel's whole point is that countries differ.

It is also the shape most flattering to the argument. An earnings-related pension paid regardless of wealth is a **risk-free real annuity**: it arrives whatever the portfolio does, which puts a floor under retirement consumption, crowds fixed income out of the financial portfolio and pushes the optimal equity share up. Nothing in this paper has tested how much of the result that floor is carrying.

Australia is the developed world's counter-example, and it differs in two ways at once. The **pension** is flat rather than earnings-related and is **means-tested**: assessable assets above a free area withdraw it on a taper until it cuts out. And the **contribution** is compulsory: the Superannuation Guarantee pays 12% of ordinary time earnings into a separate fund, on top of whatever the worker saves voluntarily. The two push in opposite directions, so the sweep crosses them rather than running one Australia-versus-America row that would confound them.

5.1 Calibration

Table 15. The Age Pension for a single retiree, March 2026 indexation, with assets-test thresholds from July 2026, against Australian average weekly ordinary time earnings of \$2,051.10 (ABS, November 2025).

Quantity	Statutory	In average earnings
Maximum single rate	\$1,200.90 a fortnight	0.293
Assets-test free area (homeowner)	\$321,500	3.014
Assets-test free area (non-homeowner)	\$600,000	5.625
Taper	\$3 a fortnight per \$1,000	0.078 a year
Cut-out (homeowner)	\$722,000	6.767
Superannuation Guarantee	12% of earnings, taxed 15% on entry	0.102 reaching the fund

Notes. Rates are held as multiples of economy-wide average earnings so the schedule travels across the panel's currencies unchanged.

The guarantee is a second contribution stream, not a larger first one. The worker still saves this paper's own 10% out of take-home pay; the employer pays 12% on top, 15% of it is taken as contributions tax, and the

remaining 10.2% is invested in the same strategy and assessed by the same means test. Total contributions are 20.2% of income against the American saver's 10%. Because both pots hold the same strategy and face no differential tax on earnings in this model, they are financially one pot and are simulated as one; the guarantee supplies 50% of everything contributed.

Working-life consumption is unchanged by the guarantee here, because its statutory incidence is on the employer. If the true incidence is on workers through lower wages — which is what most of the empirical literature finds — an Australian is paying for it in forgone pay and the comparison below is generous to them by that amount. It does not touch the certainty equivalents, because the utility window in this paper is retirement only. It does mean this is a comparison of *systems as legislated* rather than of two workers with the same lifetime resources, which is why the matched-contribution rows are there.

The taper is the other mechanism, and it is steep. A dollar of assessable assets inside the tapered band costs 7.8% of pension every year it is held. No asset in this panel earns that reliably in real terms, so inside the band the marginal return on wealth is negative — and run backwards, a retiree whose portfolio falls is met by a pension that rises. A means test is a floor and a ceiling at once, which is why it is assessed on assets as they stand, year by year, rather than settled once at retirement.

5.2 Crossing the two features

Table 16. Retirement consumption relative to the US baseline, $\gamma = 5$, 50,000 lifetimes per regime.

System	Certainty equivalent (%)	Mean consumption (%)	5th percentile (%)	All-intl over 50/50 (%)	Best strategy
US Social Security (the paper's baseline)	+0.0	+0.0	+0.0	+5.73	100% international equity
US Social Security, saving 20.2%	+23.1	+63.1	+32.8	+6.98	100% international equity
Age Pension rate, paid to everyone (no means test)	-12.9	-12.1	-10.1	+6.33	100% international equity
Australian Age Pension, saving 10%	-49.9	-34.5	-51.6	-8.06	100% bills (cash)
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	-41.0	+26.8	-26.5	-6.42	Target-date fund
Australia as legislated, non-homeowner thresholds	-34.3	+27.7	-20.3	-5.28	Target-date fund

Notes. The first three columns are levels — how much retirement consumption a system delivers. The fifth is the ranking question — which portfolio it leads its investor to hold. They are different questions and here they have different answers.

The Australian system raises the average and lowers the certainty equivalent. As legislated it delivers +26.8% mean retirement consumption against the American baseline and -41.0% certainty-equivalent consumption. Both are right, and the distance between them is the whole result: the fifth percentile is -26.5%. Compulsory saving buys a bigger portfolio and the means test takes away the floor that portfolio would otherwise sit on, so the good outcomes get better and the bad ones get worse. A criterion as averse to the lower tail as a certainty equivalent at $\gamma = 5$ prefers the annuity.

The crux is one number. Under the American schedule this investor collects a public pension worth 44% of economy-wide average earnings, every year of retirement, unconditionally and for life. The Australian maximum rate is 29% to begin with — and this investor collects a mean of 2.1%, because the guarantee has made them wealthy enough that the assets test withdraws almost all of it. They have swapped a large risk-free real annuity for a larger portfolio. That is a good trade on the average and a bad one in the tail, which is exactly what the three level columns above report.

The crossing says where each piece of that comes from. Holding the American pension and paying the Australian contribution rate is worth +23.1%: compulsory saving on its own is a large gain. Holding the contribution rate and cutting the pension to the Age Pension's flat rate, still paid to everybody, costs -12.9% and changes nothing about the ordering — it is a smaller annuity, not a different kind of one. Adding the assets test to that is what does the damage. Together they land at -41.0%: the guarantee does not buy back what the means test removes.

5.3 What it does to the ranking

The ranking does not survive the means test. All-international leads the 50/50 split by 5.73% under the American schedule and by -6.42% under the Australian one, where the best strategy becomes *Target-date fund*. This is the only place in this paper where the headline ordering fails, and it fails for a reason that has nothing to do with the return panel.

The mechanism is the floor, not the taper. The taper is certainly steep — 7.8% of assessable assets a year, more than domestic equity earns on average in this panel — and it is tempting to read the reordering off it directly. That reading does not survive checking where this household stands. Section 6.5 measures it: under either reading of the contribution rate the household is past the cut-off before the test is applied, so the taper never operates on them in the central case. It reaches them only in the left tail, once a portfolio has already fallen far enough to qualify.

What the means test removes is the American schedule's unconditional annuity, and with it the floor under bad outcomes. A retiree standing on a floor can afford the equity tail; a retiree standing on their portfolio alone cannot, and a risk-averse objective prices that difference heavily. It shows in the numbers running in opposite directions at once: against the American baseline this household's mean retirement consumption is +26.8% while its fifth percentile is -26.5%. The control is the untested row above — pay the same flat Age Pension to everyone, means-testing nothing, and the ranking is unchanged even though the level falls. Nothing about the returns has changed between any of these rows. What changed is whether anything is guaranteed.

How far it goes depends on the balance, not the country. Run the same means test on a saver with no guarantee behind them — 10% voluntary saving and nothing else — and the lead falls further, to -8.06%, with *100% bills (cash)* taking first place rather than *Target-date fund*. The poorer the saver, the deeper into the taper they sit and the further the distortion goes: at 20.2% of income the answer is a de-risking schedule, at 10% it is cash. The compulsory contribution does not remove the distortion, it softens it.

Two things follow, and they should be kept apart. The first is about this model: its investor contributes 20.2% of a rising income for 38 years and compounds it at historical real equity returns with no tax on fund earnings, which makes them far wealthier than a median Australian and puts them at the top of the taper rather than in the middle of it. The second is about the paper as a whole: every other section models a retiree whose public income is an unconditional annuity, and a retiree whose public income is withdrawn at the margin is solving a different problem. The answer here is not that Australians should hold a glide path. It is that a means test changes what the portfolio is for.

The assets test has a higher free area for retirees who do not own a home, and the model's investor owns nothing outside the portfolio. Running the non-homeowner thresholds — a free area of 5.63 times average earnings rather than 3.01 — moves the certainty equivalent to -34.3%, which is enough to matter and is reported alongside.

Table 17. Where retirees sit on the taper under the system as legislated, holding the 50/50 portfolio.

Age	Full rate	Part rate	No pension	Mean pension (× average earnings)
70	0.8%	4.2%	95.0%	0.008
80	5.2%	5.3%	89.5%	0.023
90	13.2%	3.8%	83.0%	0.045

Notes. A means test that never binds is a flat pension in disguise; one that always binds to zero is no pension at all. This table says which of the two this saver faces.

Table 18. How much of retirement the state is paying for.

System	Mean pension	Share of retirement consumption
Age Pension rate, paid to everyone (no means test)	0.449	29.7%
Australian Age Pension, saving 10%	0.063	5.6%
Australia as legislated: 10% voluntary + 12% SG, means-tested pension	0.033	1.5%
Australia as legislated, non-homeowner thresholds	0.048	2.2%

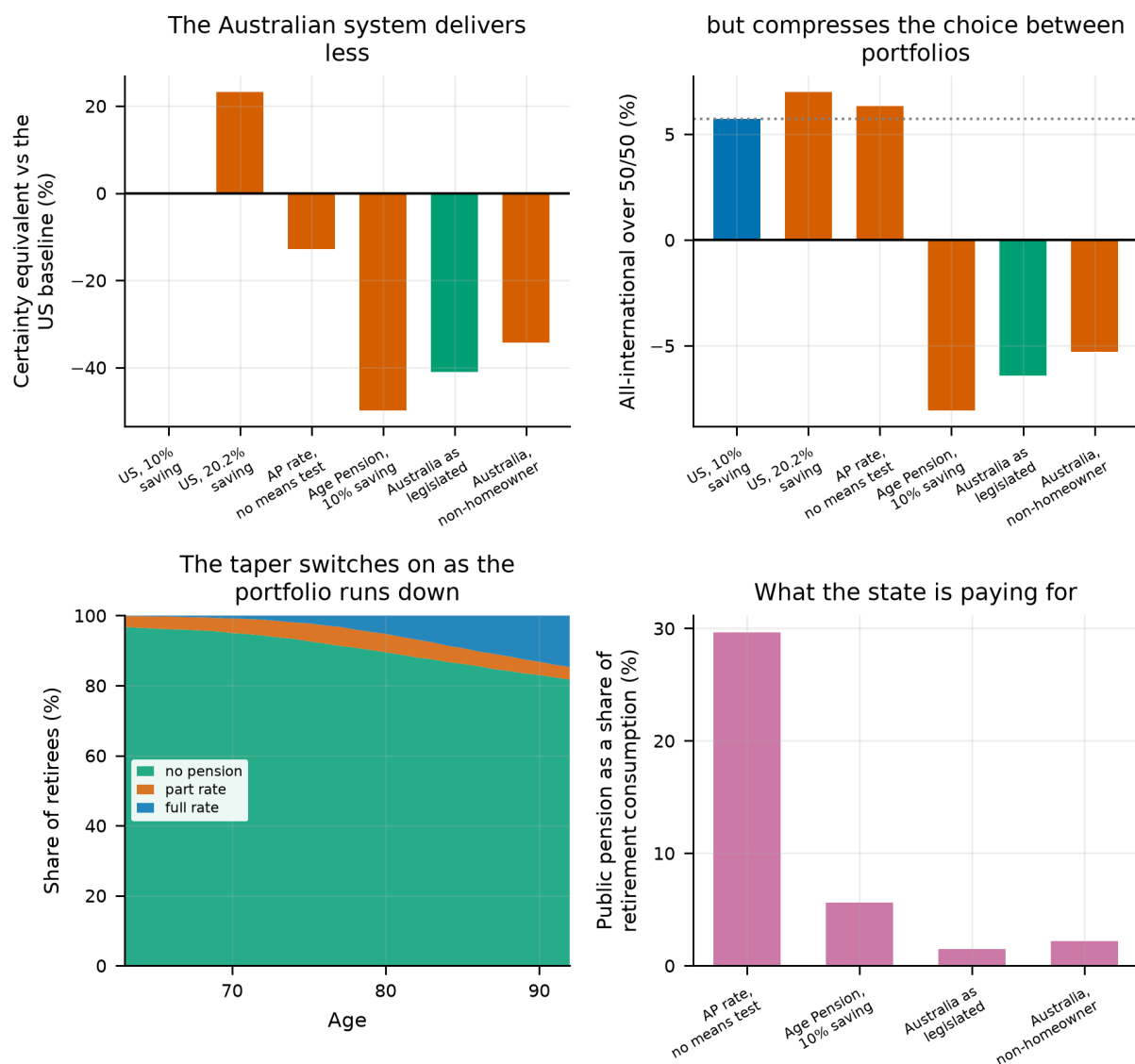


Figure 14.

Top left: retirement consumption under each regime against the American baseline. Top right: what each regime does to the choice between portfolios. Bottom left: where retirees sit on the assets-test taper as the portfolio draws down. Bottom right: how much of retirement consumption the public pension supplies.

5.4 What this changes

- **A pension schedule is not an innocent modelling choice.** Swapping one developed country's for another's moves certainty-equivalent retirement consumption by -41% while the return panel, the preferences and the portfolios stay exactly where they were. Every level quoted elsewhere in this paper is conditional on the American schedule.
- **The average and the certainty equivalent disagree, and both are true.** Compulsory saving raises mean retirement consumption +27% and lowers the fifth percentile -27%. A reader who cares about expected consumption should read the first; one who cares about the bad case should read the second. The paper's own criterion reads the second.
- **The allocation ranking reverses.** This is the only place in this paper where it does. Under the means test the best strategy becomes *Target-date fund*, and for a saver poorer than this one it goes further still. Every other section models a retiree whose public income arrives regardless of what they own; that is the

assumption the ranking rests on.

- **What is not modelled:** the income test, the family home's exemption from the assets test — the largest single feature of the real system — the tax on fund earnings in accumulation, and every other tax in this paper. Nor is an annuity, which is what the American schedule is and what an Australian retiree would have to buy to match it.

6. What a Year of Retirement Is Worth

Section 31 of the companion study solves the withdrawal rule and the allocation together, and finds that the third leg of a retirement plan — the date — cannot be solved at all. Left free it runs to the oldest age on the grid and stays there, because nothing in this model charges anyone for the years they spend working. Another year is more contributions, a shorter drawdown and a larger benefit, at no cost whatever. This section supplies the missing side of that ledger.

Rather than invent a disutility of labour in utils — a quantity nothing here can calibrate — the cost is written as a *consumption equivalent*. Let L be the number such that a retired year at consumption c is worth exactly as much as a working year at $L \cdot c$. Working years are then evaluated at c/L and retired years left alone, so $L = 1$ charges nothing and reproduces every other result in this paper exactly.

The aggregation here runs from age 25, not from the retirement date. A retirement-window objective cannot price a retirement date: it charges the investor for the years they worked and credits them nothing for the ones they did not. These certainty equivalents are therefore not comparable with those elsewhere in the paper.

6.1 The pension you claim early

There is a larger thing to fix first. Every other section holds the retirement date fixed, so the benefit starts on the same birthday for every strategy and its start date cancels. It does not cancel here: left alone, this model pays whoever stops work at fifty a full unreduced pension for forty-three years. That is not a preference for leisure, it is a gift, and it is worth more than any plausible value of leisure.

Real systems reduce a benefit claimed early and raise one deferred, by roughly enough to leave its expected present value alone. The multiplier that does that exactly is the ratio of annuity factors, $A(\text{reference})/A(\text{age})$, derived from this model's own Gompertz survival curve and discount factor rather than taken from a statute.

Table 19. The actuarially fair claiming adjustment, measured against age 63.

Retire at	Benefit multiplier	Adjustment (%)	Per year from the reference (%)
50	0.432	-56.8	-6.25
52	0.485	-51.5	-6.36
55	0.582	-41.8	-6.54
58	0.706	-29.4	-6.73
60	0.807	-19.3	-6.88
62	0.930	-7.0	-7.04
63	1.000	+0.0	—
65	1.164	+16.4	-7.30
67	1.366	+36.6	-7.50
70	1.771	+77.1	-7.84

Notes. Fair by construction in this model rather than approximately fair in someone else's: it is solved from the same survival law Section 17 of the companion study uses.

That works out at 6.3% to 7.8% a year away from 63. The US schedule reduces a benefit by about 6.7% for each of the first three years claimed early and raises it about 8% for each year deferred past full retirement age. Nothing here was fitted to that; the agreement is a check on the survival law, not a coincidence to lean on.

The claiming rule decides the answer before leisure gets a say. Unadjusted, the best date is 52 even when working costs nothing at all. Adjusted, it is 60. Everything below is the adjusted arm.

6.2 When to stop

Table 20. The best retirement date at each value of leisure, $\gamma = 5$.

A retired year is worth	Retire at	Lifetime CEC	Runner-up	Margin (%)
+0%	60	0.7668	58	0.89
+5%	60	0.7345	58	0.39
+10%	58	0.7045	60	0.02
+15%	58	0.6787	60	0.37
+20%	58	0.6543	60	0.66
+25%	58	0.6313	60	0.91
+30%	58	0.6095	60	1.12
+40%	58	0.5695	55	0.28
+50%	55	0.5365	58	0.47
+75%	55	0.4685	52	1.16
+100%	52	0.4153	55	0.34

Notes. An optimum sitting on either end of the age grid would be the grid's answer rather than the model's; the grid runs from 50 to 70 for that reason.

Pricing the cost of working gives the date an interior optimum. It runs from 60 when a retired year is worth no more than a working one to 52 at the top of the grid. Section 31 of the companion study's corner is gone — not because the model changed its mind, but because both sides of the decision are now priced.

Weighting by survival pulls it earlier still, at 2 of the 11 values of leisure tested and by up to 3 years. Deferring is a bet that you will be there to collect, and the Gompertz law of Section 17 of the companion study prices that bet where a certain ninety-third birthday cannot.

6.3 The break-even, which is the number to carry

An optimal date depends on a calibration nobody can hand you. A break-even does not. For each date earlier than the one an investor would choose if working cost nothing, this is the value of leisure at which it becomes worthwhile.

Table 21. What each year of earlier retirement costs to justify, against age 60.

Retire at	Years earlier	Leisure must be worth	which is a consumption drop of
58	2	9.7%	9%
55	5	31.3%	24%
52	8	60.4%	38%
50	10	81.2%	45%

Notes. The reference is the date that wins when working costs nothing, so what is priced is the earlier retirement rather than the model's own lean toward it.

That turns the question into one a reader can settle for themselves. Stopping 2 years early is a claim that a year of your own time is worth at least 10% of a year's consumption — about the size of the consumption drop actually

observed at retirement. Stopping 10 years early is a much larger claim: 81%. Whether either is true, this paper does not say.

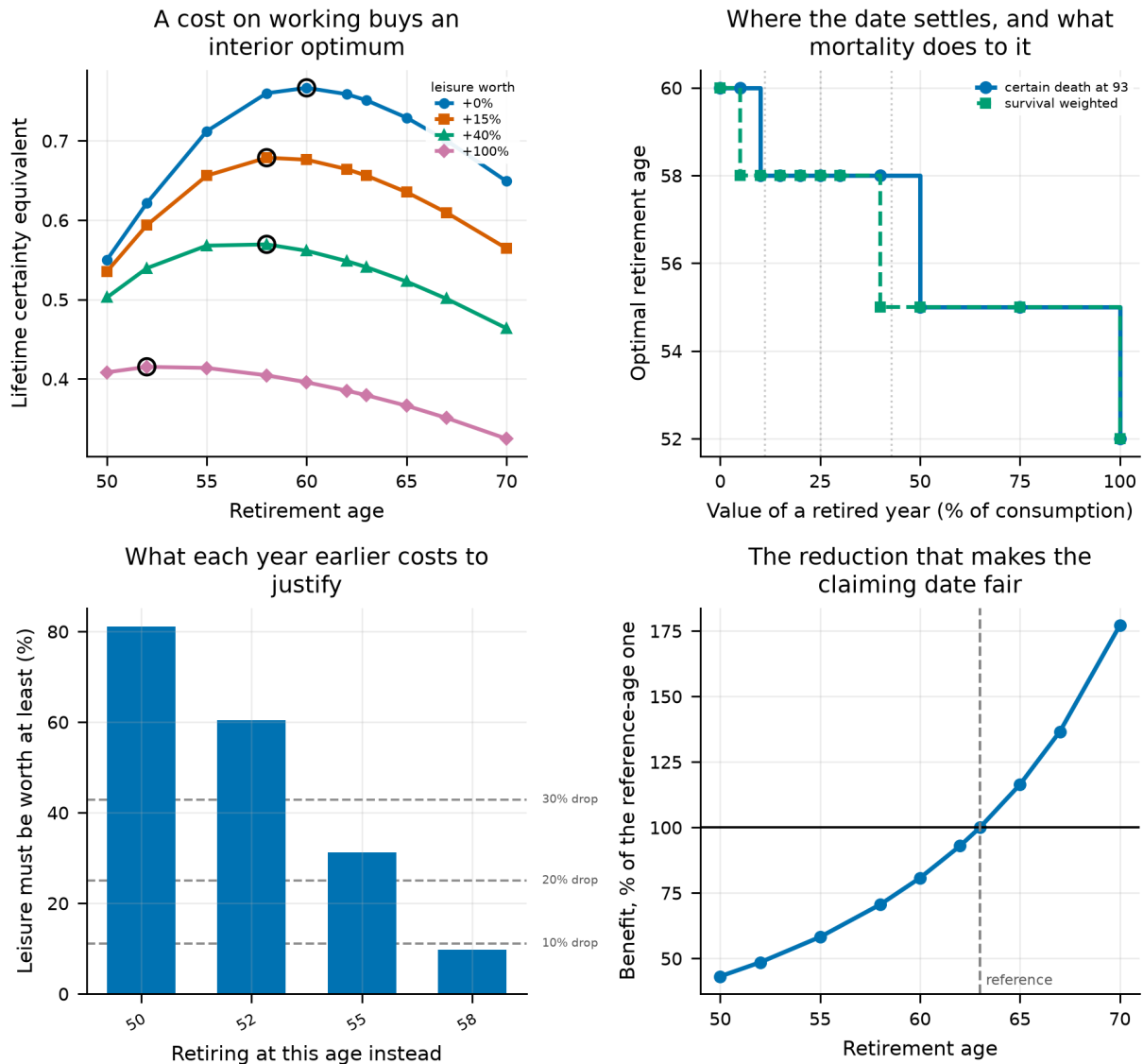


Figure 15.

Top left: lifetime certainty equivalent against the retirement date, one line per value of leisure, with each line's maximum circled — a cost on working buys an interior optimum. Top right: how that optimum moves, with and without survival weighting. Bottom left: the break-even value of leisure for each earlier date, against the consumption drops observed at retirement. Bottom right: the claiming adjustment that makes the date of the pension worth nothing either way.

6.4 The same question under Australia's pension

Everything above pays an American pension: earnings-related, payable from the day work stops, and adjusted for the age it starts at. Australia's is shaped differently in two ways that pull against each other. The Age Pension is payable at 67 however early somebody stopped, so there is no claiming choice to adjust and no bridge — an Australian retiring at 55 funds twelve years alone before a cent arrives. And the Superannuation Guarantee puts 12% of earnings into the portfolio on top of voluntary saving, which is what pays for crossing that bridge. Running the pension alone and then the pension with the guarantee separates the two.

Table 22. The retirement date under each pension, $y = 5$.

Pension system	Best date, leisure free	at the top	Earlier dates any leisure justifies	Nearest earlier date costs	per year
US social security	60	52	4 of 4	9.7%	4.9%
Age Pension only	67	60	6 of 8	16.0%	8.0%
Age Pension + super guarantee	62	52	5 of 5	4.2%	2.1%

Notes. Break-evens are measured against each system's own zero-leisure date, so they price the earlier retirement rather than the system's own lean.

Australia's pension puts the date 7 years later. On the same voluntary saving the best date moves from 60 under the American schedule to 67. Nothing about the investor changed. But the two systems differ in *two* ways at once, so that number belongs to the pair and not to either of them — which is what Section 6.5 is for.

Compulsory saving takes some of it back. Adding the Superannuation Guarantee moves the date from 67 to 62, recovering 5 of the 7 between the two systems. That one is a single change — the guarantee is on or off and nothing else moves — so unlike the row above it can be read as a cause.

Net of both, an Australian's best date is 2 years later than an American's at the same voluntary saving (62 against 60). And it moves the *slope* as well as the start. A year earlier than the best date costs an Australian 2.1% of lifetime certainty-equivalent consumption against an American's 4.9%, a factor of 0.44. The Australian system does not shift one trade-off later so much as pose a softer one: the date is later *and* each year taken off it is cheaper. Both follow from the same gate — with the pension fixed at 67, a year of retirement bought before then is funded entirely from the portfolio, so it costs more than a year bought against a benefit that moves with it.

This arm needs a caveat the others do not. A retiree who exhausts the portfolio before the pension age receives, here, a means-tested payment at 60% of the full rate, standing in for the working-age safety net. At zero it would be literally nothing, and a CRRRA aggregator punishes a single year of that without limit — one path in ten thousand at the consumption floor is enough to decide a certainty equivalent on its own. That floor is a judgement, and the earliest dates are sensitive to it in a way the later ones are not. Nor is preservation age modelled: this portfolio can be drawn at 50, where Australian super cannot.

6.5 Which of the two differences does the work

The comparison above changes two things at once. Australia's pension starts on a fixed birthday rather than when work stops, *and* it is worked out by a means test rather than from a career of earnings. Attributing the gap to either without crossing them is guesswork, so this is the 2×2: each feature alone, then both, against the same baseline.

Table 23. Each feature of Australia's pension against the American baseline, $y = 5$.

What changes	Best date	Years later	CEC	vs baseline
when the benefit starts	58	-2	0.7754	+0.0086
how the benefit is worked out	67	+7	0.6292	-0.1376
both together	67	+7	0.6292	-0.1376
interaction	—	+2	—	-0.0086

Notes. The baseline is the American schedule: earnings-related, payable from retirement, actuarially reduced for claiming early.

It is how the benefit is worked out, not when it starts. Changing only the start date moves the best date by -2 years. Changing only the formula — a means-tested pension replacing an earnings-related one, still payable the day work stops — moves it +7. The eligibility gate moves the date the *opposite* way to the system it belongs to, and the reason is worth stating: a pension payable at a fixed age is not reduced for stopping early, whereas an actuarially adjusted one is. The gate removes a bridge and a penalty at once, and on this panel the penalty is worth more than the bridge.

The two do not add. The interaction is +2 years against a joint +7, or 29% of it — which is what one feature disarming the other looks like. Once the means test has taken the pension away, the birthday it would have arrived on stops mattering.

And the formula does so much because this household is not poor enough to be paid. The assets test begins withdrawing the pension at 3.0× average earnings and reaches zero at 6.8×. The household in the 2×2 above — which holds contributions fixed, and so saves only what this paper's own saver does — has a median of 13.8×, 2.0 times the cut-off, with 83% of them past it. Against a full rate worth 29% of career-average income, what is actually paid averages 3.9%; the American schedule, means-tested against nothing, pays 43%. So this is only partly a comparison of pension *designs*. It is substantially a comparison between a household that receives a pension and the same household receiving almost none.

That household is not, however, an Australian one. Holding contributions fixed is what separates the pension's timing from its formula, and it is the wrong household for any sentence about Australia, which makes its contributions compulsory. Carrying the Superannuation Guarantee moves the median from 2.0 to 4.7 times the cut-off, the share past it from 83% to 98%, and what the pension actually replaces from 3.9% down to 1.9% of career income. The guarantee does not carry this household *through* the means test; it carries them 2.3 times further past it.

This settles what the taper can and cannot be blamed for. It is steep enough to act as a wealth tax — every dollar inside the band costs more pension a year than domestic equity earns in this panel — but a rate that is never reached cannot be the mechanism behind anything. Under either reading the household is past the cut-off before the test is applied, and the taper touches them only in the left tail, once a portfolio has already fallen far enough to qualify. What separates the two systems here is therefore not the withdrawal rate on the pension. It is that one system pays this household an unconditional annuity and the other pays them almost nothing, and the annuity is worth most exactly where the portfolio is worth least.

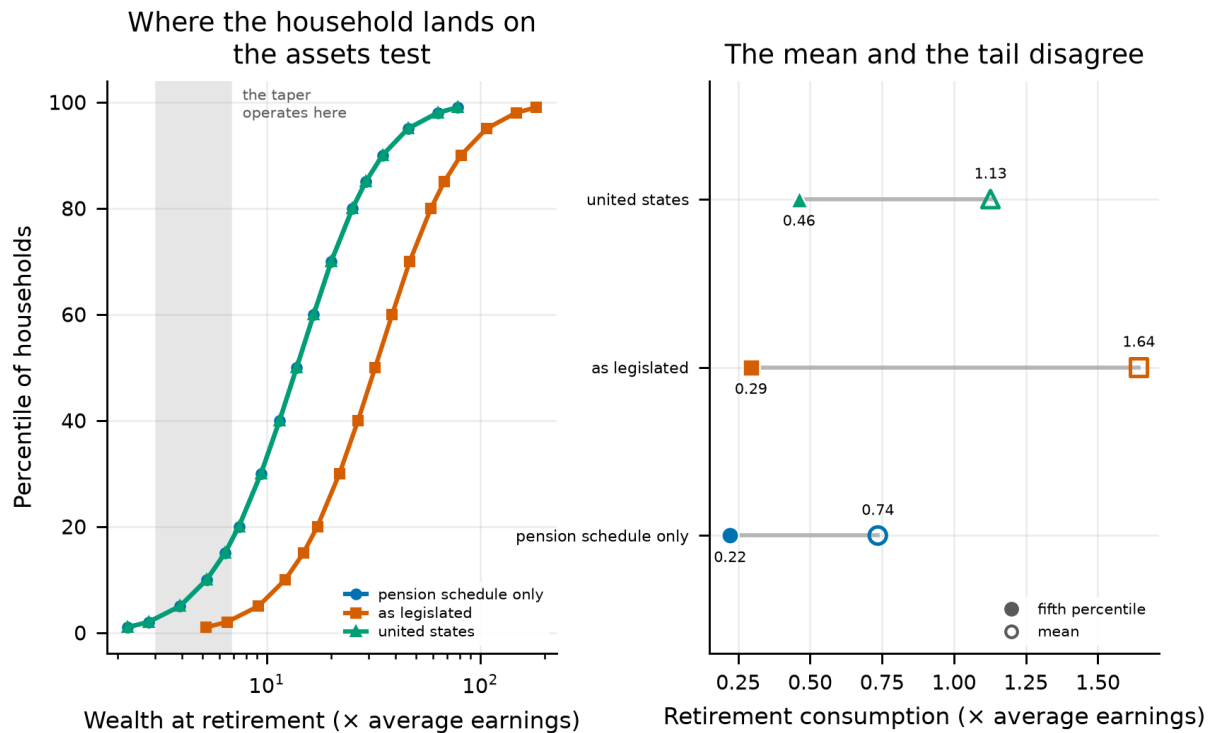


Figure 16.

What the assets test does, and does not, explain. Left, the distribution of wealth at retirement for each household against the band the taper operates in — on a log scale, because the distance involved is multiplicative. Both Australian households sit almost entirely to the right of the band, so the taper is answered before it is applied. Right, the mean and the fifth percentile of retirement consumption for each. The two run in opposite directions: the household with compulsory saving has the highest mean of the three and nearly the lowest tail, which is the gap the certainty equivalent is pricing.

6.6 The withdrawal rule is half the comparison

Everything above spends by this project's own rule: a fixed real amount set as a share of the portfolio at retirement. That rule cannot answer the question a pension is *for*. It sets the standard of living from the portfolio, so a larger balance spends more rather than lasting longer, and the pension is spent on top of the withdrawal rather than funding part of it. The alternative fixes the target instead — a set share of pre-retirement income for life, the pension netted off, the portfolio paying the rest.

Table 24. Each pension system under each withdrawal rule, $\gamma = 5$, retiring at 63.

Rule	Pension system	CEC	Ruin (%)	Mean c	5th pct c
fixed_real_rule	US social security	1.0494	11.7	1.729	0.708
replace_50	US social security	0.6057	2.5	0.852	0.438
replace_75	US social security	0.6759	11.7	1.136	0.454
replace_100	US social security	0.7578	24.8	1.373	0.500
fixed_real_rule	Age Pension only	0.4226	11.7	1.129	0.341
replace_50	Age Pension only	0.3417	10.3	0.755	0.255
replace_75	Age Pension only	0.4917	26.1	1.014	0.379
replace_100	Age Pension only	0.5782	41.0	1.190	0.449
fixed_real_rule	Age Pension + super guarantee	0.7242	11.7	2.520	0.452
replace_50	Age Pension + super guarantee	0.3354	1.9	0.797	0.250
replace_75	Age Pension + super guarantee	0.4991	6.8	1.162	0.374
replace_100	Age Pension + super guarantee	0.6403	13.4	1.485	0.449

Notes. The replacement rules target a share of average income over the final working years, held in real terms.

Under the portfolio rule all three systems ruin at the same rate, to the last decimal. That is an artefact, not a finding, and it is worth naming because this project has leaned on that rule throughout: withdrawals set as a share of wealth at retirement scale with the portfolio, so doubling the balance doubles the spending and exhausts it in the same year. Ruin under that rule is a property of the return sequence alone, and says nothing about how much was saved.

Hold the target still and the Superannuation Guarantee finally shows up where it should. At 100% replacement the Australian household runs out 11.4 percentage points less often than the American one. Compulsory saving buys safety — but only under a rule that permits safety to be bought.

And the certainty equivalent still favours the American schedule at every target. Both facts hold at once, and the combination is the result: the Australian household runs out *less often* and is *poorer when it does*. What remains once the portfolio is gone is a flat means-tested pension rather than an earnings-related one, and a CRRA aggregator at this risk aversion prices the depth of the floor above the frequency of reaching it. A reader who cares about the probability of running out and a reader who cares about the worst case will rank these two systems differently, and neither is misreading the table.

6.7 The rule decides which system wins

Every row above spends by a rule that cannot adapt: a fixed real amount, set as a share of wealth at retirement and then held there for thirty years. Section 7 finds that rule is not the best available, and that the best one is an amortisation rule — the balance divided by an annuity factor — dialled by the real return it assumes. That dial is swept here rather than set, because the two systems arrive at retirement with very different portfolios and there is no reason their best assumption should be the same.

It is not the same, and the comparison does not survive the change. The United States is best served by assuming 4% and Australia by 7% — a household with a larger portfolio and almost no pension has more to convert into consumption and less reason to hold back. Scored at each system's own best assumption the gap is +11.3%, against -31.0% under the fixed real rule the rest of this paper spends by. The sign changes. Australia is behind under one withdrawal rule and ahead under another, on the same returns, the same pension schedules and the same contributions, with the crossing at an assumed 3%.

Which qualifies Section 6.6's finding rather than overturning it. The mechanism there was the floor: strip the unconditional annuity away and the portfolio's own worst outcomes become the whole of retirement. That is still what is happening — but the floor does not have to come from a pension. An amortisation rule divides by the years remaining, so it cannot run the balance to zero, and ruin falls to nothing at every assumption on this grid. A household that provides its own floor out of a larger portfolio is no longer paying for the pension it does not receive. The Australian result reported above is therefore a statement about a fixed real withdrawal rule at least as much as it is a statement about the Australian pension.

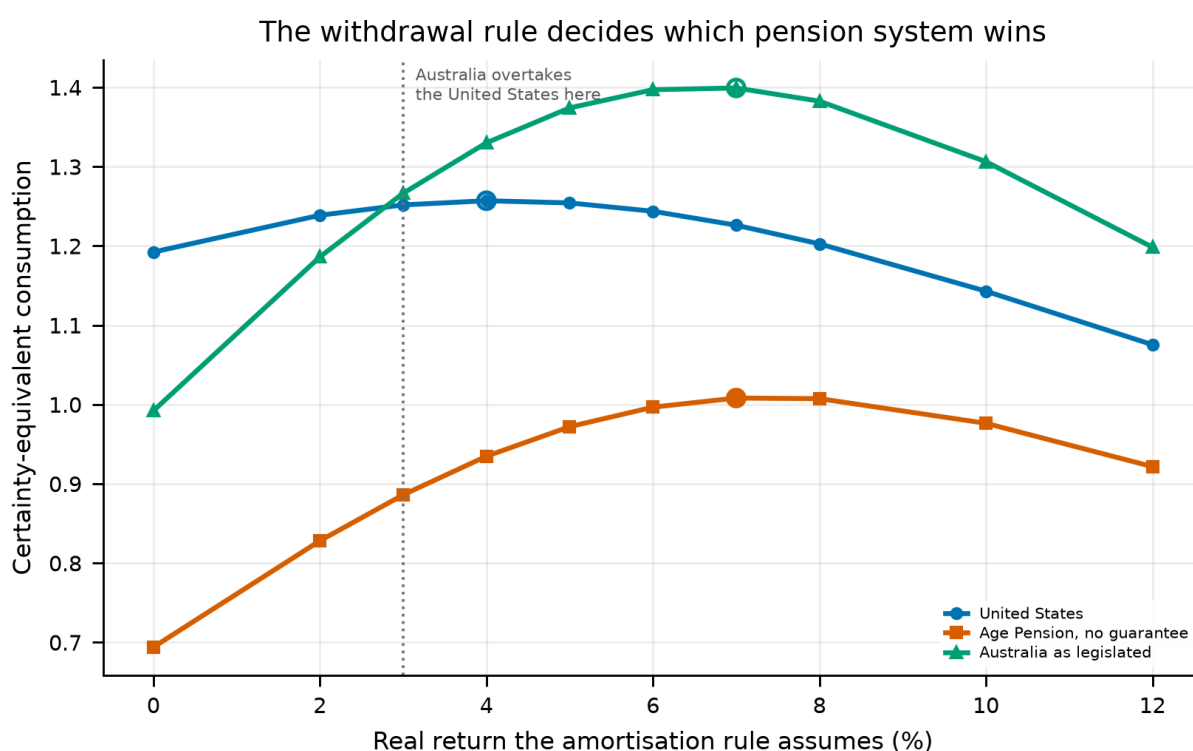


Figure 17.

Certainty-equivalent consumption against the real return an amortisation rule assumes, for each pension system. One axis and one measure: the question is which curve is higher and where, and rings mark each system's own optimum. The curves cross, which is the point — the sign of the gap between two pension systems is not a property of the pensions alone.

6.8 What this changes

- **The retirement date is not unpriceable — it was unpriced.** Charging for the years spent working turns Section 31 of the companion study's corner into an interior optimum between 52 and 60, depending on what a year is worth.
- **The claiming rule matters more than leisure does.** An unreduced pension starting whenever work stops is worth more than any plausible value of leisure and would have decided the answer on its own. That is worth stating for its own sake: a lifecycle model that lets the retirement date move must price the pension's start date, or it is not measuring preferences at all.

- **The output is a break-even, not a recommendation.** The value of your own time is not something this panel of returns can measure. What it can do is say what you must believe to justify a given date, and leave the believing to you.
- **What is not modelled:** partial retirement, which is what most people actually do; any change in the value of leisure with age or health, when both plainly change; an earliest claiming age, so the adjusted benefit here can start at fifty where no real system would pay it; and the possibility that work is worth something positive — purpose, company, structure — which would push the date later and which this parameterisation cannot represent, since L is bounded below at one.

7. The Rule When the Horizon Is Not Known

Every comparison of withdrawal rules so far — Section 30 of the companion study's sweep, Section 31 of the companion study's joint optimisation — scores a retirement that ends at ninety-three with certainty. That is not neutral between rules. A rule that amortises to a fixed date is handed the answer, because dividing by the years remaining is very nearly optimal when the years remaining are a known constant. A rule that hedges longevity pays a premium against a risk the model has switched off. And ruin counts as failure a portfolio exhausted at ninety-one for an investor who most likely died some years before.

Section 17 of the companion study re-weights the aggregation by a Gompertz survival curve and finds it does not change which *allocation* wins. It also says, in terms, that the treatment is only approximate for the horizon-based rules, “which would themselves change if they knew the mortality table”. This section is that unfinished sentence: the rule and the rate are re-solved alongside the allocation, on one objective.

The grid is 2,280 combinations of allocation, rule and rate, each scored twice on the *same* simulated lifetimes — once over the fixed horizon, once survival-weighted. No path is re-drawn between the two, so a difference between the scores is the aggregation and nothing else. On this calibration the expected age at death is nan, so a retiree stopping at 63 should plan for nan years against the nan assumed throughout. The model was not merely uncertain about the horizon; it was assuming one substantially longer than the median retiree gets.

Table 25. The best combination under each aggregation, $\gamma = 5$.

Scored over	Rule	Equity	Domestic	CEC
fixed horizon	gompertz	100%	10%	1.3844
survival-weighted	amortisation (6% assumed return)	100%	10%	1.4232

Notes. Both winners derive their level from a planning horizon and so set no rate of their own.

The horizon changes the rule and nothing else. Freeing the allocation to be re-chosen for a real lifespan is worth +0.00% and freeing the rate +0.00%; freeing the rule is worth +2.81%, which is the whole of the +2.81% available from re-choosing all three, with an interaction of +0.00%. The three decisions do not interact here, and that is worth stating as plainly as an interaction would have been.

The rate optimum is interior: the best rate-setting rule, *constant_percent*, wants 8.0% inside a grid running 3.0% to 12.0%, so the curve turns over rather than stopping and the number is a peak rather than the end of the grid.

The assumed return has an interior optimum too. *amortisation* (6% assumed return) wants 6% inside a grid running 0% to 12%, so over-assuming does begin to cost and the sweep can see where it starts. This is a dial the rule comparison in Section 30 of the companion study never swept: the amortisation rule enters that menu at assumed returns chosen by hand, and the best of those is not the same thing as an optimum.

The rate a rule wants spans 3.5 percentage points. *constant percent* wants 8.0% and *constant real* wants 4.5%, across 9 rules that set a rate at all. The tempting explanation is that the rules which cannot run out want the high

rates, and it is nearly right but not right: endowment (light smoothing) wants 8.0% and *can* deplete, tying the top of the list. What sets the rate is how far spending scales with the portfolio rather than the ability to run out as such — a lightly smoothed endowment rule is most of a percentage of balance, and is priced like one.

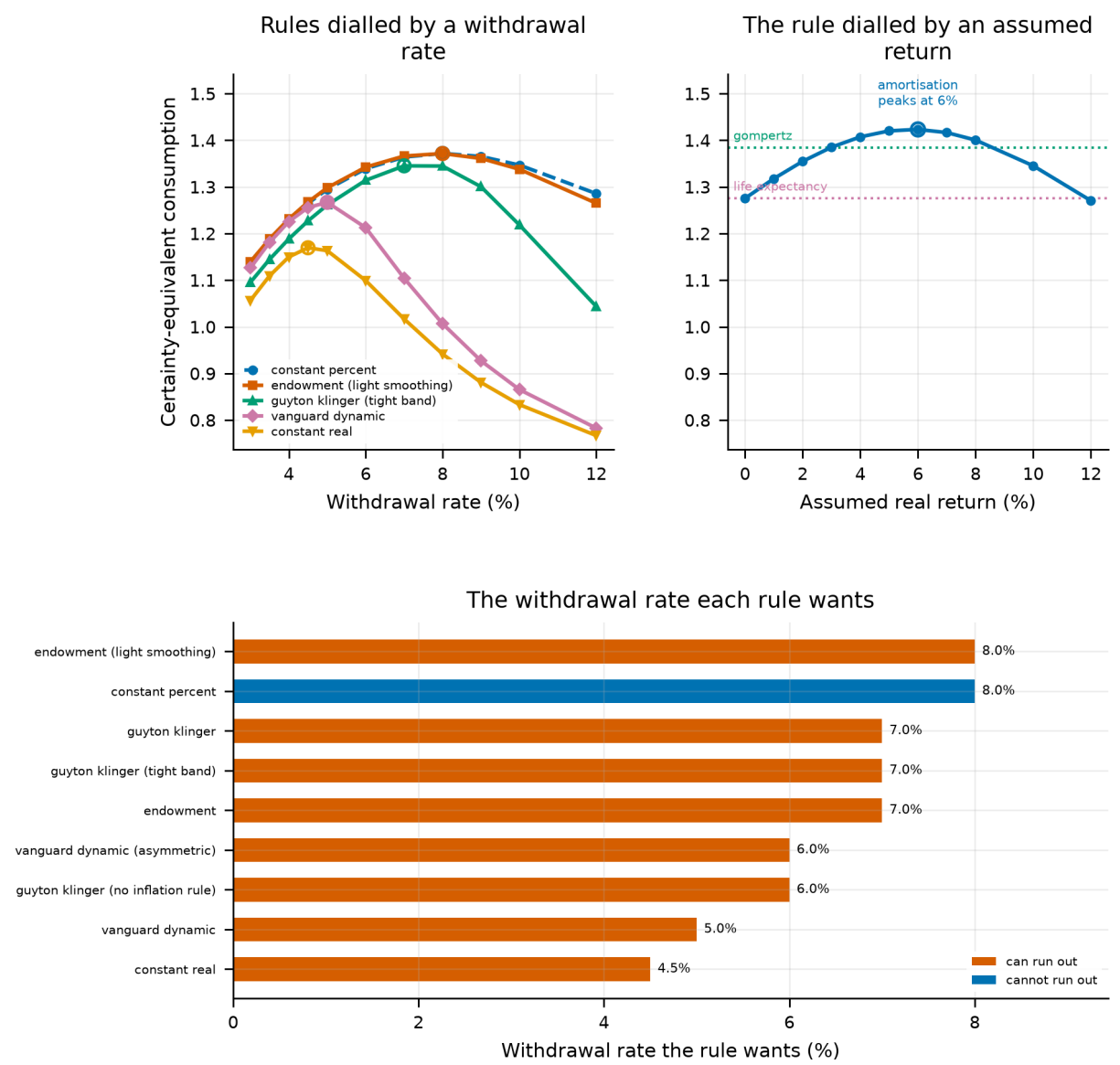


Figure 18. Where each dial that sets a spending level actually peaks. Top left, the certainty equivalent against the withdrawal rate for the five rules that take one; rings mark where each curve turns over, and solid lines can run out where dashed cannot. Top right, the same against the assumed real return for the rule dialled by one instead, with the two rules that set no dial at all drawn as the constant levels they are. A withdrawal rate and an assumed return are different measures, so they get an x-axis each; the vertical scale is shared between them, which is what lets the two peaks be compared by eye rather than by reading numbers off two ranges. Below, the withdrawal rate each rule wants.

7.1 Which rules a real lifespan promotes

Table 26. Every rule at its own best rate and allocation under the objective being ranked.

Rule	Fixed horizon	Real lifespan	Places gained
amortisation (6% assumed return)	5	1	+4
amortisation (5% assumed return)	3	2	+1
amortisation (7% assumed return)	7	3	+4

Rule	Fixed horizon	Real lifespan	Places gained
amortisation (4% assumed return)	2	4	-2
amortisation (8% assumed return)	11	5	+6
amortisation (3% assumed return)	4	6	-2
gompertz	1	7	-6
constant percent	8	8	+0
endowment (light smoothing)	9	9	+0
endowment	13	10	+3
amortisation (2% assumed return)	6	11	-5
amortisation (10% assumed return)	19	12	+7
guyton klinger (tight band)	14	13	+1
guyton klinger	15	14	+1
guyton klinger (no inflation rule)	18	15	+3
amortisation (1% assumed return)	10	16	-6
gompertz (+5y buffer)	12	17	-5
vanguard dynamic (asymmetric)	22	18	+4
amortisation (0% assumed return)	16	19	-3
life expectancy	16	19	-3
amortisation (12% assumed return)	23	21	+2
vanguard dynamic	21	22	-1
life expectancy (+5y buffer)	20	23	-3
constant real	24	24	+0

Notes. Scoring each rule at a setting chosen to suit a different horizon would confound the rule with the rate.

Part of the reason is the spending level rather than the mortality table. Rank each rule by the share of the balance it draws in the first retirement year — a number every rule has, however it arrives at one — and that order matches the survival-weighted ranking with a rank correlation of +0.48, against +0.06 under the fixed horizon. The contrast is the finding rather than either level, and at +0.48 it is a contributing cause and not the whole of one. The winning rule has never heard of a survival curve: the best rule that reads one ranks 7. So the answer to “should the withdrawal rule know how long you are likely to live?” is that it does not need to — it needs to spend as though the answer were shorter than thirty years, and there is more than one way to arrange that.

7.2 The ruin number everyone quotes

Ruin roughly halves. On the best rule that can actually run out — endowment — the fixed horizon reports 0.9% against the real lifespan's 0.5%, and across all 1,449 combinations that can deplete the median ratio is 2.0. Nothing about any portfolio changed. What changed is that the earlier number counts as a failure a portfolio exhausted at ninety-one for somebody who, on this survival curve, most likely died before reaching it. Every ruin probability elsewhere in this paper carries the same overstatement.

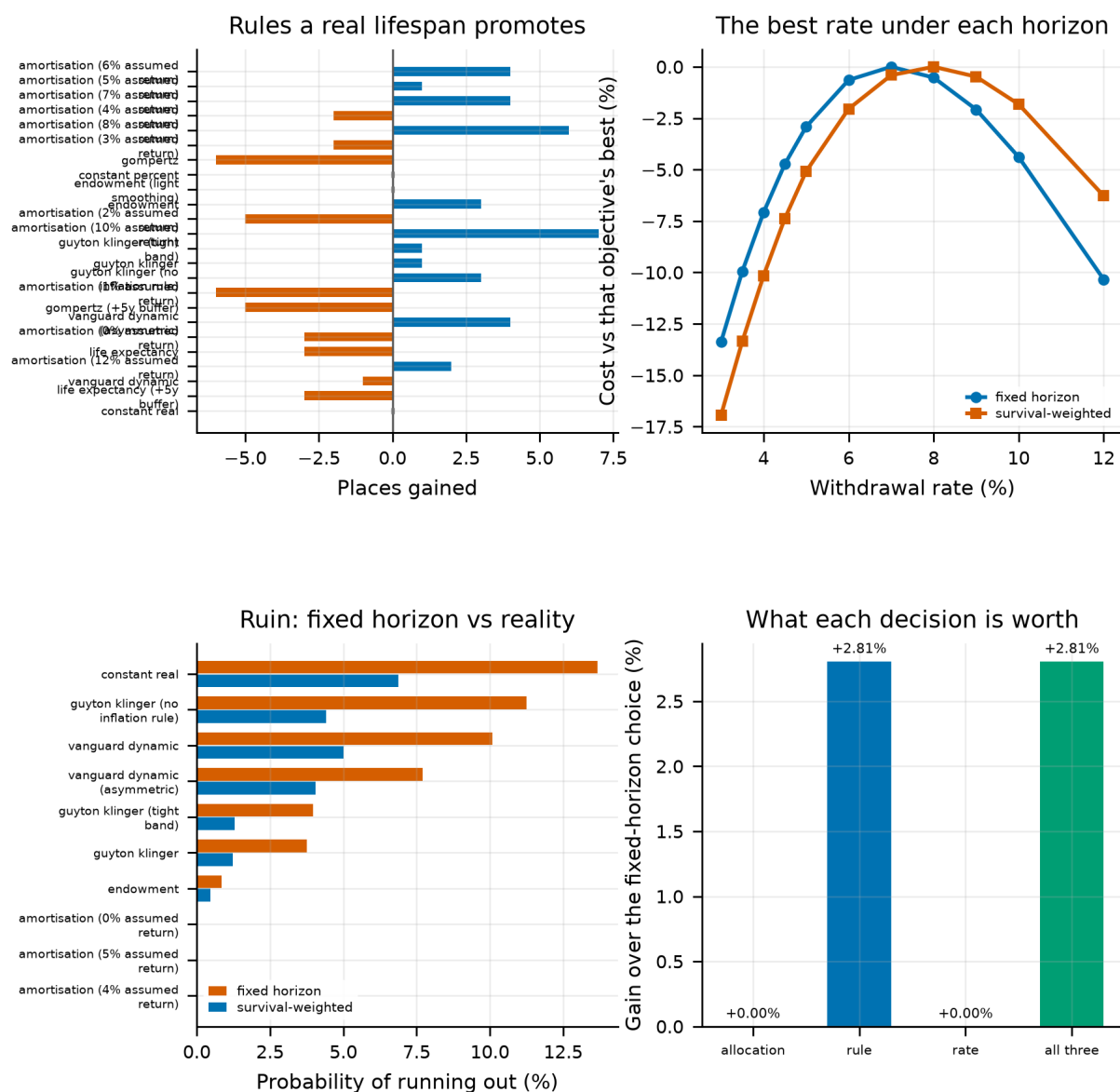


Figure 19.

What an uncertain lifespan does to the rule, the rate and the risk. Top left, how far each rule moves in the ranking; top right, where the best rate sits under each objective, each scaled to its own best so only the shape is compared; bottom left, ruin under both aggregations; bottom right, what re-choosing each decision is worth.

What is not modelled: annuities, which are the direct hedge for the risk this section prices and would dominate part of the grid if they were in it; a rule that conditions on realised health rather than on the table it started with; and couples, where the relevant horizon is the second death and the distribution is a different shape. The Gompertz curve is a two-parameter model rather than a life table, and Section 17 of the companion study sweeps those parameters; the grid here is held at that section's central calibration.

8. The Tax Each System Actually Charges

Every result so far is tax-free, and until Section 6 that was harmless. A charge every strategy pays alike cancels out of a comparison between strategies, so the ranking of portfolios is unchanged by leaving it out. Comparing two *countries* is where the argument fails, because what differs between the American and Australian retirement systems is precisely which income is taxed, and when.

The asymmetry is the reason it could not be reasoned about. Australia charges early and lightly — on fund earnings through accumulation, and far less than the 15% headline implies — and then not at all: superannuation

drawn after 60 from a taxed fund is not merely tax-free but *not assessable income*, so it cannot affect the tax on anything else. The Age Pension is assessable, but the seniors and pensioners offset lifts the effective threshold above the full single rate, so a pensioner living on the pension pays nothing. The United States charges late, progressively, and interactively: social security is taxable on a sliding scale whose thresholds have not been indexed since 1993, and a withdrawal from a traditional account is ordinary income *and* counts toward the provisional income deciding how much of the benefit is taxed.

8.1 The torpedo, measured

At the withdrawal this household actually makes, the American retiree faces a marginal rate of 22.2% while sitting in a 12% bracket — a factor of 1.85, which is the arithmetic ceiling: every dollar drawn drags up to eighty-five cents of benefit into the tax base behind it. The Australian retiree faces no such thing, because the withdrawal that would do the dragging is not assessable income.

The shape matters as much as the level. The marginal rate rises, then *falls* as the 85% inclusion completes, then rises again into the next bracket — so it is not monotone in income, and a retiree who looked up their bracket would be wrong about the cost of their next withdrawal across most of the range this household occupies.

8.2 What it does to the comparison

Table 27. Each system tax-free and under its own schedule, $\gamma = 5$.

Pension system	Tax regime	CEC	Ruin (%)	Tax paid (% of gross)
US social security	No tax	1.0409	11.9	0.0
US social security	Roth	1.0409	11.9	0.0
US social security	Traditional	1.0620	11.9	9.4
Age Pension only	No tax	0.4269	11.9	0.0
Age Pension only	Australia, super after 60	0.4269	11.9	0.0
Age Pension + super guarantee	No tax	0.7223	11.9	0.0
Age Pension + super guarantee	Australia, super after 60	0.7181	11.9	0.0

Notes. The two American rows are the honest pair: a Roth is what every earlier section implicitly assumed, and a traditional account is what superannuation deserves to be compared with, since both take contributions from pre-tax earnings.

Tax is worth +2.0% of certainty-equivalent consumption to an American saving in a traditional account, and costs 0.6% to an Australian. To an American saving in a Roth it costs nothing at all: with no other assessable income the benefit alone stays below thresholds frozen since 1993, and the torpedo needs other income to fire. The gap between the two countries moves from -30.6% to -32.4%, and the ranking holds.

The gap widens, by 1.8 percentage points — and which side moved matters, because a gap can widen from either end. 78% of the movement is the American arm *gaining* rather than the Australian arm losing: a pre-tax contribution grows by the tax not paid on it, and what is owed later is owed on a lower income, so the deferral is worth more than the retirement tax costs. Australia's charge is small at both ends — nothing in retirement, and very little during accumulation once unrealised gains, the capital-gains discount and franking credits are counted, which the next subsection sets out.

8.3 What the Australian fund actually pays

The statutory rate on fund earnings is 15%, and charging that against the return would overstate the charge by more than an order of magnitude. Three things stand between the headline and what is paid, and on this portfolio they very nearly cancel it. **Unrealised gains are not income**: a fund that holds owes nothing on appreciation, and because earnings in the retirement phase are exempt outright, a gain carried across that boundary is never taxed at all. **A realised gain held beyond a year is discounted by a third**, so the rate on it is ten per cent. And **a franked dividend carries a credit worth more than the liability** — Section 15 of the companion study derives it on this same panel — so a domestic dividend is a refund that subsidises the tax on the international sleeve rather than adding to it.

Together they leave a drag of 0.073% a year against the 1.20% the statutory rate on the return would imply, a factor of 16. At a domestic share of 50% the income side comes out positive: the credit more than covers the fund's tax on the rest of the portfolio.

Table 28. The Australian charge by how much the fund is made to sell.

Sold each year	Drag a year	CEC	Ruin (%)	Against holding to the pension phase (%)
0.00%	+0.1282%	0.7297	11.9	+0.0
3.35%	-0.0727%	0.7181	11.9	-1.6
10.00%	-0.4718%	0.6956	11.9	-4.7
25.00%	-1.3718%	0.6483	11.9	-11.1
50.00%	-2.8718%	0.5788	11.9	-20.7

Notes. Holding until the pension phase is the reference. Section 22 of the companion study measures 3.35% a year for this strategy, all of it forced by drift rather than chosen.

Holding until the pension phase is not a fanciful case — it is what an index fund left alone does. At the 3.35% a year Section 22 of the companion study measures for this strategy, the charge costs 1.6% of the certainty equivalent.

8.4 The whole comparison in one chart

The sections above each change one thing, but report it against its own reference. This walks the same ground in a single line, from the American schedule to the Australian one, turning on one feature at each step so that a step's height is that feature's contribution and nothing else's.

Table 29. From US social security to Australia as legislated, $\gamma = 5$.

Turning on	CEC	Change	Share of the gap (%)
US social security	1.0409	—	—
+ pension age 67	0.9574	-0.0835	+26
+ means test	0.4269	-0.5305	+164
+ super guarantee	0.7223	+0.2954	-92
+ Australian tax	0.7181	-0.0042	+1

Notes. The retirement date is held fixed here. Section 6 lets it move, and the pension age reads differently when it can.

One step is the answer. Turning on means test moves the certainty equivalent by -0.530; the smallest step, Australian tax, moves it -0.004 — a ratio of 126. Whatever separates these two systems, it is not the tax, and it is not mainly the pension age either.

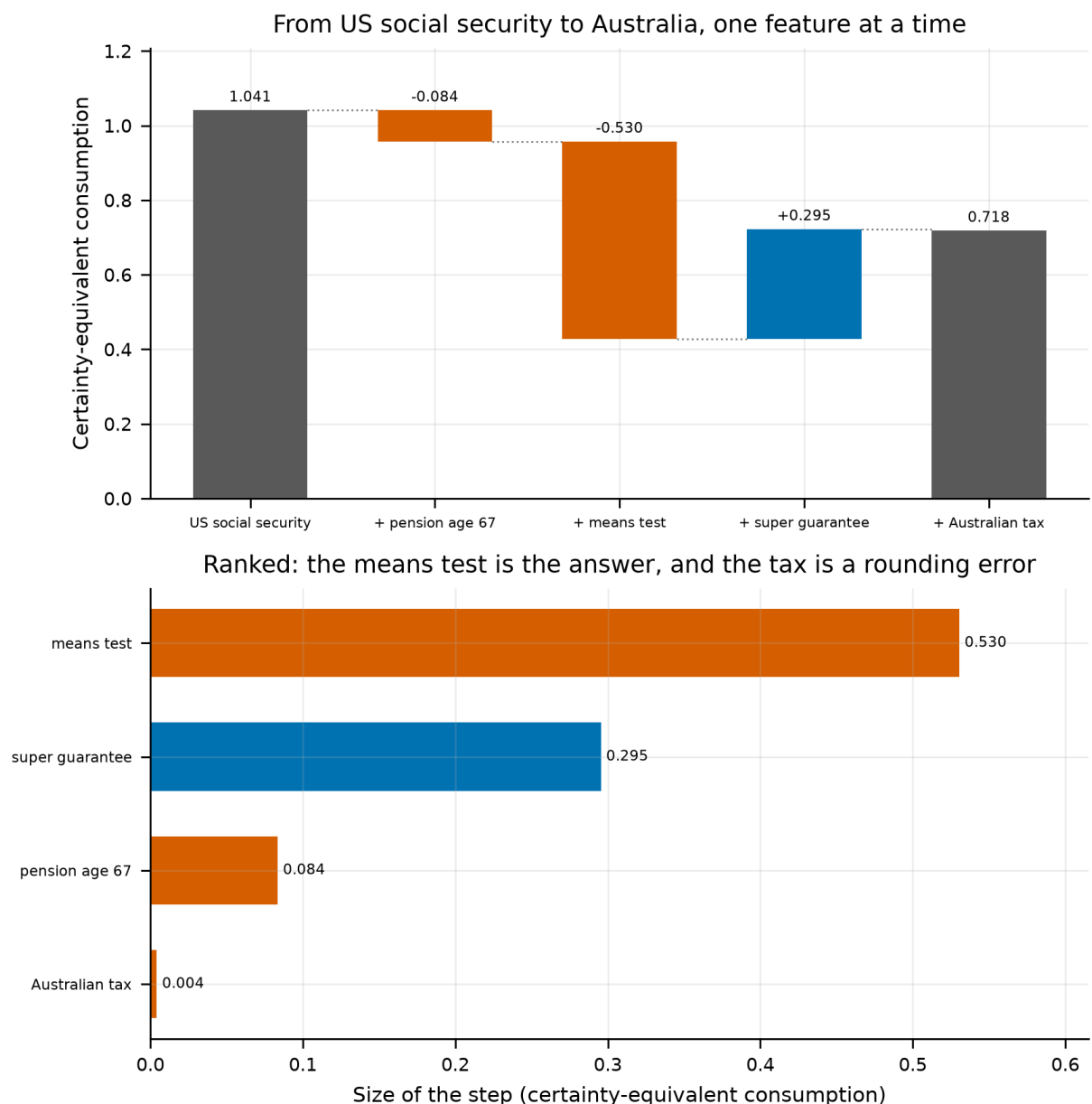


Figure 20.

US social security to Australia as legislated, one feature at a time. Endpoints sit on the floor and are levels; the steps between them float, so a bar's height is the change that feature makes. The lower panel ranks the same steps by size. The retirement date is held fixed at the reference age.

What is still missing, in rough order of how much it would move this: the Australian marginal tax on earnings held outside super, since voluntary saving is treated as a Roth in both countries to isolate the retirement wrapper; capital gains tax and its one-third discount inside a fund, which would push the 15% on fund earnings down; US state income taxes, which would push the American charge up; and any filing status but single.

9. Limitations and Threats to Validity

This section is deliberately long. A replication that reports only the ways in which it succeeded is not much use.

9.1 The cross-section is sixteen countries

This is the largest weakness in the paper, and it is a weakness we chose deliberately. The developed-market universe runs to 38 markets. We cover 16, because the other 22 have no recorded return series in any openly licensed source. Section 2.6 audits what we do use; Section 2.6.1 reports what the excluded countries have and why it is not enough.

Every statement in this paper about international diversification is therefore a statement about an average over 15 other developed markets, weighted by history length and drawn jointly with the domestic leg. That is a real limit on how far the mechanism generalises, and in particular the panel is entirely advanced-economy and entirely survivor: markets that closed and never reopened are not in the source database, so the tail this paper measures is the tail of markets that came back.

Section 8 of the companion study quantifies the part of this that is quantifiable. No single country carries the headline, and the ranking is stable across every window of history tested — but the delete-one jackknife puts a standard error on the headline gap that is wide enough to matter, and a reader should carry that interval to every other number in the paper. Nothing here is estimated more precisely than sixteen countries allow, however many decimal places the simulation resolves. What Section 8 of the companion study cannot address is the survivorship in the sample frame itself: deleting a country that is present says nothing about a country that was never there.

The last five years of the series are separately flagged as unverified in Section 2.6, on a variance test that every country in the sample fails in the same direction.

9.2 What the bootstrap cannot represent

- **No valuation conditioning.** Blocks are drawn without regard to the starting dividend yield or price-earnings ratio, so a simulated lifetime beginning at a market peak is statistically identical to one beginning at a trough. If long-horizon returns are predictable from valuations, the dispersion here is too wide and the mean too high.
- **No regime structure.** Block resampling preserves local persistence but not the possibility that the whole return-generating process changes. Secular declines in real yields, changes in the corporate tax base and the shift toward intangible capital are all invisible.
- **System-level survivorship.** Every country in the panel still had a functioning market in 2020. The set of histories in which that is false has zero probability under this sampler.
- **Blocks respect gaps, and gaps are informative.** A country with a wartime hole contributes fewer long blocks, so the sampler is mildly biased away from exactly the disrupted histories that motivate the exercise. We quantify this rather than correct it.
- **The realised backtest selects on windows, not just on countries.** Section 7 of the companion study avoids the sampler entirely by running the lifetimes history actually served, but the filter that makes that possible — an unbroken run the length of a working life and a retirement — removes every window that straddles a catastrophe rather than following one. The markets whose equity was destroyed therefore contribute only the lifetimes that begin at the floor of the recovery. That biases the comparison toward domestic equity, so the lead reported there is conservative in direction; it also means the spread across countries in that section is not the spread a randomly chosen saver faced.

9.3 What the lifecycle model omits

- **One tax, and only one.** Section 14 of the companion study prices foreign dividend withholding, because it is the tax that falls unequally on the two legs this paper compares and is therefore the one that can change the answer — and it does, inside the statutory range. Every other tax is absent: returns are otherwise pre-tax, withdrawals untaxed, and there is no distinction between a taxable and a tax-deferred account. The asymmetric treatment of capital gains and income would change the ranking of spending rules in Section 30 of the companion study more than it would change the allocation ranking; dividend imputation, which refunds corporate tax to domestic shareholders in Australia and New Zealand, would push the same way as withholding and is not modelled either.
- **Every return is gross.** No fee is charged anywhere in the baseline. Section 13 of the companion study prices the omission rather than repairing it, by finding the fee differential that would undo the headline, and Section 22 of the companion study does the same for the cost of trading. Both leave the baseline itself gross, so every level quoted outside those two sections is a number no investor receives.
- **No housing in the baseline.** For most households the primary residence is the largest asset and the mortgage the largest liability, and both interact with inflation in ways the financial portfolio does not. Sections 24 of the companion study and 25 of the companion study add housing and leverage against it to the opportunity set, but every other result in this paper is computed without them. Section 2.6.2 audits the 1,805 country-years of observed housing returns behind that, and explains why an appraisal-smoothed index cannot be used as written.
- **No economy-wide wage growth.** The income profile is an age effect only. Section 2.6.3 measures what that leaves out — a median 1.41% a year across 18 countries, which compounds to 1.68× over a career — and finds the bias runs against our conclusion, because less human capital weakens rather than strengthens the case for early equity.
- **No annuities.** A real annuity is the natural competitor to a withdrawal rule and is absent entirely.
- **One country's pension behind sixteen countries' returns.** Every result outside Section 5 pays the US primary-insurance-amount schedule, and that is not a neutral choice: an earnings-related benefit paid regardless of wealth is a risk-free real annuity, which puts a floor under retirement consumption and flatters equity. Section 5 is the one place this is tested, against Australia's means-tested Age Pension and compulsory Superannuation Guarantee, and it moves the level further than any parameter swept anywhere else in this paper. The ranking survives there only because the guarantee lifts the model's saver clear of the means test; a saver inside the taper band gets a different answer, and nothing else in this paper speaks to them.
- **No disutility of labour in the baseline.** Working is free everywhere except Section 6, which charges for the years spent at it and turns the retirement date from a corner into an interior optimum. Everywhere else — including the timing results of Section 28 of the companion study, which for that reason require a matched comparison — the model cannot say when someone *should* retire, only what conditioning the date is worth. What is still missing is any value on the *kind* of work: a year is a year here, whoever spends it doing whatever.
- **A fixed horizon everywhere but two sections.** Death arrives at 93 with certainty in every result except Sections 17 of the companion study and 7. The first re-weights the headline under four Gompertz survival laws and finds the ranking unmoved, which is a narrower finding than it sounds: it re-weights a policy that does not itself respond to longevity, so a small answer is partly built in. The second closes that gap by re-solving the rule and the rate alongside the allocation under a survival curve, and finds the winner is a rule that never reads a mortality table at all. What is still missing is an annuity, which is where the value of longevity risk actually sits.

- **No behavioural constraints.** Every rule here assumes the investor executes it. Section 27.6 of the companion study prices the cost of a constrained contribution but nothing prices the probability that a saver abandons an all-equity portfolio in the middle of a 60% drawdown.

9.4 Specification sensitivities we know about

Three modelling choices are load-bearing and a reader should know where they bite.

- **The bequest shift.** Without De Nardi's κ , a single path with exactly zero terminal wealth sends the certainty equivalent to negative infinity at $\gamma \geq 1$. The value of κ is not identified by anything in the data and it changes the ranking of spending rules.
- **The evaluation window.** Allocation comparisons use the retirement window because working-life consumption is identical across strategies by construction; policy comparisons that change working-life consumption use the whole lifetime. Mixing the two would produce incoherent comparisons and we flag every switch.
- **The working-income floor.** Retirement carries a consumption floor through the progressive benefit schedule and working life, at zero, carries none. That asymmetry cancels in the allocation comparisons and does not cancel in the timing ones, where leaving it uncorrected inflates the value of retiring early by roughly a factor of two.

9.5 What the new searches assume

- **The simplex search is a coordinate search.** Section 20 of the companion study reports restarts from three corners and they agree, but coordinate ascent offers no guarantee of a global optimum in a 204-dimensional space and none is claimed.
- **The opportunity set is still narrow.** Section 24 of the companion study adds housing as a fifth asset, but there is no credit, no commodities and no inflation-linked bond, and an inflation-linked bond in particular would change the fixed-income result of Section 20 of the companion study more than any refinement within the sleeves that are present.
- **The valuation conditioning applies to the first block only.** A lifetime is a chain of calendar windows and only the first is a starting condition; the rest are the future, which no investor chooses. Section 11 of the companion study therefore measures the effect of the opening decade's valuation on a sixty-eight-year outcome, diluted by everything that follows. A design that made the whole chain valuation-dependent would report a larger effect and would be assuming a great deal more.
- **The de-smoothing of housing is itself a model.** Section 24 of the companion study inverts a first-order filter, which is the standard correction and not necessarily the right one: if house price indices carry higher-order smoothing, the true volatility is higher than the correction restores and housing is worth less than reported. The direction of that error is known even though its size is not.
- **The mortgage is rebalanced annually.** Section 25 of the companion study redraws the loan every year at no cost to hit a target loan-to-value. Real mortgages amortise on a fixed schedule, cost several percent of the property to refinance, and are called on missed payments rather than on a drifting ratio. The section reports the value of the leverage, not a financing plan, and it prices no mortgage insurance and no tax deductibility in either direction.
- **Housing is priced as an index, not as a house.** The asset in Section 24 of the companion study is a liquid, continuously rebalanced, nationally diversified claim on the housing stock. A single leveraged owner-occupied property shares almost none of those properties, and nothing in this paper speaks to it.

- **Leverage is modelled with limited liability** and a floating borrowing rate, rebalanced annually. Real margin lending liquidates at a threshold rather than at zero and reprices continuously, both of which would make borrowing less attractive than Section 21 of the companion study finds it.
- **No borrowing constraint binds by quantity.** The model lets an investor borrow three times their financial capital at a spread; no lender would extend that against a retirement account.

9.6 Statistical caveats

Certainty equivalents are reported without standard errors. Under common random numbers the *differences* between policies are far more precisely estimated than their levels, which is what the comparisons rely on, but a reader wanting a confidence interval on any single level will not find one here. The optimisers are coordinate-ascent procedures on a grid: exact per coordinate under common random numbers, but with no guarantee of a global optimum in the joint space. Multiple restarts are reported where it matters.

- **The cohort interval rests on 16 clusters.** The confidence interval in Section 7 of the companion study is a cluster bootstrap over countries, and a percentile interval built on 16 clusters has poorer coverage than its width suggests. The cohorts are also unequally distributed across those clusters, so the pooled mean is implicitly weighted by how much unbroken history a country happens to have. Both readings are reported there; the country-equal-weighted one is the conservative one.
- **One split is one experiment.** Section 23 of the companion study cuts the record once and solves on either side. The asymmetry it finds is real in the sense that it is what the data say, but a single cut cannot separate a schedule that was overfitted from a world that changed, and with one experiment there is no distribution to judge the asymmetry against. It should be read as a caution rather than as a test with a size.

10. Conclusion

The case for a fixed all-equity lifecycle portfolio is a case about an investor with a guaranteed real income underneath it. Give the same investor the same returns and an asset-tested pension instead, and the lead of 5.73% becomes -6.42% and the de-risking glide path — the design the literature spends its length arguing against — wins.

The reason is worth separating from the obvious one. It is tempting to attribute the reversal to the means test's taper, which is steep enough to act as a wealth tax. It cannot be that: this household is past the cut-off before the test applies. It is the loss of the floor. Paying the same pension to everyone, with no test at all, leaves the ordering intact (+6.33%) even though it pays this household less than the American schedule does. Withdrawing a pension at the margin and paying a smaller one are different interventions, and only the first reorders portfolios.

The second finding qualifies the first and should travel with it. A withdrawal rule that cannot deplete the portfolio supplies the floor the pension no longer does, and the ordering reverses again. So the reversal is not a fact about Australia. It is a fact about any retiree whose income in the bad states depends on the portfolio itself — which includes a saver under an asset test, and excludes one drawing on an annuity, a defined-benefit scheme, or a rule that divides by the years remaining.

Two implications follow for default design. A plan sponsor choosing a glide path is implicitly choosing it against a pension schedule, and the same fund is not the right default in a means-tested system and an earnings-related one. And the drawdown default matters as much as the portfolio default: in a system without a floor, supplying one through the withdrawal rule recovers what the pension no longer provides.

We state these as implications of a calibrated model. It carries no behavioural constraints, prices no annuity, and — as Section 9 sets out — rests on a sixteen-country panel whose jackknife interval is wide enough that the direction of these results is far better established than their size.

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